

A Categorical Viability-Weighted Curvature Framework: From Autopoietic Sets to Non-Abelian Holonomy

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Abstract. A single categorical construction is developed that unifies viability-constrained adaptation, reflexively autocatalytic food-generated (RAF) reaction networks, the Fisher–Rao information-geometric structure of belief updates, completely-positive trace-preserving (CPTP) quantum channels with Zeno-type measurement schedules, and non-abelian holonomy of policy loops under the structure group $\mathrm{CO}(n-1)$. The construction rests on the Stratified Autopoietic Viability Geometric System (SAVGS), a five-component object (continuous control manifold, policy fiber, open probability simplex, strict viability margin, endogenous maintenance graph) whose geometry is a stratified principal $\mathrm{CO}(n-1)$ -bundle; the viability-weighted curvature κ_V is derived as the positive part of the directional derivative of a Bregman divergence evaluated at the algorithmic rate-distortion distance, an intrinsically single-string quantity that eliminates the deterministic-versus-random- coding gap of classical rate-distortion theory. The construction assembles seven prior arcs of domain unification into one endofunctor on the optic category **Optic**($\mathcal{C}$) of Riley; the resulting composition theorem (Theorem 7.4) makes the unification object a well-defined fixed point whose existence reduces to an empirical contraction-rate question, decoupled from definitional rhetoric. A Bregman-divergence Noether correspondence (Proposition 5.1) supplies the affine-invariant structure required for the gauge invariance of κ_V . A seven-claim falsification hierarchy (Definition 6.1) is operationalized in the minimal $n = 3$ prototype (abelian, $\mathfrak{so}(2)$) and in the $n = 4$ non-abelian prototype ($\mathfrak{so}(3)$, $[\iota_z, \iota_x] = \iota_y$); the heavy-tail index of the fatigue noise is stress-tested across the Student- t tail-parameter axis. The main proposition (Proposition 15.1) states that on smooth constant-active-set strata, Fisher-minimal constraint-preserving adaptation defines a stratified connection whose viability-weighted curvature predicts leading-order policy hysteresis. All seven claims are confirmed within their stated tolerances.

Keywords: viability theory, optic category, RAF sets, Fisher–Rao metric, CPTP channels, quantum Zeno effect, non-abelian holonomy, heavy-tailed noise, Banach fixed point, Bregman divergence, algorithmic rate-distortion, Noether correspondence.

1 Introduction

The notion that adaptation of a system to viability constraints can be unified across cognitive, biological, and physical domains has a long informal history in autopoiesis, viability theory [1], and information geometry [2, 3], but the precise categorical content of the unification has remained in flux. The dominant informal pattern is that viability-preserving systems, reflexively autocatalytic food-generated (RAF) reaction networks, predictive (Bayesian) belief updates, and quantum measurement schedules each admit a description in terms of a cost functional C whose stabilizer $\mathrm{Stab}(C)$ defines the structure group of the category of admissible updates. This pattern, however, has been expressed as a series of bridge-like correspondences without a single composition construction making the unification object a fixed point of a well-defined endofunctor.

This article assembles seven arcs of prior domain-unification work into a single endofunctor T on the optic category **Optic**($\mathcal{C}$) of Riley [4] and Brunerie et al. [5], where $\mathcal{C}$ is taken to be a category with finite limits. The composition $T = \mathcal{O}_7 \circ \cdots \circ \mathcal{O}_1$ of the seven optics is well-defined, associative, and unital; the unification object is the fixed point of T whenever T is contractive (Theorem 7.4). The framework rests on the Stratified Autopoietic Viability Geometric System

(SAVGS, Section 3), a five-component object whose geometry is a stratified principal $\text{CO}(n-1)$ -bundle; on this object, the viability-weighted curvature κ_V is well-defined, gauge-invariant, and falsifiable.

The viability-weighted curvature κ_V is the leading-order operational form of the cost functional C , defined on policy loops in the agent parameter space. It is constructed (Section 4) as the positive part of the directional derivative of a Bregman divergence h_α evaluated at the algorithmic rate-distortion distance $\text{dist}_D(x)$, a single-string quantity that eliminates the deterministic-versus-random-coding gap of classical rate-distortion theory. For a circular loop γ_a of amplitude a in a radially symmetric viability landscape V with maximum $V_{\max}$ at the origin, the operational form reduces to

$$\kappa_V(a) = \frac{1}{V_{\max}} \langle V_{\max} - V \circ \gamma_a \rangle = a^2, \quad (1)$$

independent of dimension; the structure group $\text{Stab}(\kappa_V)$ is then the endogenous structure group of the system and is computed, not chosen.

A Bregman-divergence Noether correspondence (Section 5, Proposition 5.1) supplies the affine-invariant structure required for the gauge invariance of κ_V : under a one-parameter affine transformation on the dual coordinate, the Bregman divergence is invariant, and Noether’s theorem yields a conserved current. The correspondence is a theorem with explicit, checkable preconditions: the distortion measure and the source prior must both be Bregman divergences. A system that violates this precondition refutes the correspondence.

The principal contributions are summarized as follows.

1. The SAVGS framework (Definition 3.1), a five-component stratified object that resolves the type confusions, gauge ambiguities, and missing premises of prior formulations. The geometry is a stratified principal $\text{CO}(n-1)$ -bundle, and κ_V is computed on this object. The Fisher-minimal horizontal lift on each constant-active-set stratum is given in closed KKT form (Proposition 3.4); the projected differential inclusion at constraint-switching boundaries is given in Remark 3.5; the small-loop viability–holonomy theorem (Theorem 3.6) gives the leading-order ε^2 expansion of the endpoint margin and the sufficient condition for endpoint viability; the closure-aware curvature κ_{closure} of Definition 3.12 isolates autopoietic failure from behavioral hysteresis.
2. A single composition theorem (Theorem 7.4) for the seven optics in **Optic**($\mathcal{C}$), replacing the rhetorical “bridge-rung” construction of prior work. The unification object is the fixed point of T , well-defined up to canonical isomorphism whenever T is contractive.
3. The algorithmic rate-distortion distance dist_D (Definition 4.1) and the derivation of κ_V from a Bregman divergence evaluated at dist_D (Proposition 4.4), supplying the deterministic single-string content required for the per-trajectory falsification protocol.
4. A Bregman-divergence Noether correspondence (Proposition 5.1) that turns the prior Noether analogy into a theorem with checkable preconditions.
5. A seven-claim falsification hierarchy (Definition 6.1) on the operational form κ_V , with five derivative claims (A–E) in the $n = 3$ prototype, an $n \geq 4$ non-abelian extension (Claim F), and a CPTP-Zeno quantum lift (Claim G).
6. Numerical confirmation of all seven claims within their stated tolerances, including heavy-tail index stress-testing of Claim D across the Student- t tail-parameter axis (Section 10) and generalization of A–E to the $n = 4$ non-abelian regime (Section 11).
7. An inverse-limit construction of the directed system of RAFs in **Optic**(**Set**), yielding the maximal RAF $R_{\max}$ as the colimit (Construction 14.1).
8. A numerical Banach contraction argument for T over $X = [0, 1]^d$ across a (d, k) robustness grid of 375 configurations, demonstrating that the unification object exists by Banach’s theorem [6] throughout the extended setting (Section 13).

9. A main proposition (Proposition 15.1) stating the joint thesis in its strongest defensible form: on smooth constant-active- set strata, Fisher-minimal constraint-preserving adaptation defines a stratified connection whose viability-weighted curvature predicts leading-order policy hysteresis; whether that holonomy is fatal depends on viability margins, along-path disturbances, and regeneration of internal maintenance machinery.

The remainder of the article is organized as follows. Section 2 fixes the categorical and operational preliminaries. Section 3 constructs the SAVGS framework. Section 4 introduces the algorithmic rate-distortion distance and derives κ_V . Section 5 states the Bregman-Noether correspondence. Section 6 states the falsification hierarchy. Section 7 proves the composition theorem and lists the seven optics. Section 8 reports the foundational test results (Claims F and G). Section 9 operationalizes A–E in the $n = 3$ prototype. Section 10 stress-tests Claim D’s heavy-tail index. Section 11 generalizes A–E to $n = 4$. Section 12 treats the CPTP-Zeno lift in detail (Claim G). Section 13 presents the T-iteration contraction results. Section 14 presents the inverse-limit RAF construction. Section 15 states the main proposition. Section 16 discusses limitations, implications, and future directions. Section 17 concludes.

2 Preliminaries

Definition 2.1 (Viability-weighted curvature, operational form). Let M be a smooth state manifold, $V : M \rightarrow \mathbb{R}$ a smooth viability function with maximum $V_{\max} = \max_M V$, and $\gamma_a : [0, 1] \rightarrow M$ a smooth policy loop of amplitude a . The *per-loop viability-weighted curvature* is

$$\kappa_V(\gamma_a) = \frac{1}{V_{\max}} \int_0^1 (V_{\max} - V \circ \gamma_a(t)) dt. \quad (2)$$

For the radially symmetric viability $V(x_1, \dots, x_{n-1}) = 1 - \sum_i x_i^2$ and the circular loop $\gamma_a(t) = (a \cos 2\pi t, a \sin 2\pi t, 0, \dots, 0)$, the operational form $\kappa_V(a) = a^2$ follows. The full construction of κ_V as a directional derivative of a Bregman divergence evaluated at dist_D is given in Section 4.

Definition 2.2 (Structure group). The *endogenous structure group* of the agent is $G_C := \text{Stab}(C)$, the stabilizer of the cost functional C in the automorphism group of the admissible update maps. For the Fisher–Rao metric on the n -dimensional probability simplex (Chentsov’s theorem [2, 3]), the stabilizer is $G_C = \text{CO}(n-1) = \mathbb{R}_+ \ltimes \text{SO}(n-1)$ with Lie algebra $\mathfrak{so}(n-1)$, which is abelian for $n = 3$ ($\mathfrak{so}(2)$ is 1-dimensional) and non-abelian for $n \geq 4$ ($\mathfrak{so}(3)$ is 3-dimensional with $[\mathfrak{l}_z, \mathfrak{l}_x] = \mathfrak{l}_y$).

Remark 2.3 (Physical meaning of the $n = 3 \rightarrow n = 4$ transition). At $n = 3$ the policy simplex is two-dimensional, the Fisher–Rao stabilizer is $\text{CO}(2) = \mathbb{R}_+ \ltimes \text{SO}(2)$, and the Lie algebra $\mathfrak{so}(2)$ is one-dimensional, generated by the single infinitesimal rotation about the origin. Two rotations about the same origin necessarily commute (they lie in the same one-parameter subgroup), so the holonomy of a sequence of loops is path-independent up to homotopy, and the path-ordering in the parallel-transport exponential collapses to ordinary commutative multiplication. The non-abelian signature (Claim F) is then unobservable: the commutator $[R_1, R_2] = 0$ identically, regardless of the rotation amplitudes. At $n = 4$ the policy simplex is three-dimensional, the stabilizer is $\text{CO}(3) = \mathbb{R}_+ \ltimes \text{SO}(3)$, and $\mathfrak{so}(3)$ is three-dimensional with three independent generators $\{\mathfrak{l}_x, \mathfrak{l}_y, \mathfrak{l}_z\}$ satisfying $[\mathfrak{l}_z, \mathfrak{l}_x] = \mathfrak{l}_y$ etc. Two rotations about *distinct* axes (e.g. $R_z(\alpha_1)R_x(\alpha_2)$ vs $R_x(\alpha_2)R_z(\alpha_1)$) no longer commute, and the small-angle commutator is $[R_1, R_2] \approx \alpha_1\alpha_2[\mathfrak{l}_i, \mathfrak{l}_j]$ with Frobenius norm $\sqrt{2}\alpha_1\alpha_2$ (each $\mathfrak{so}(3)$ basis element has Frobenius norm $\sqrt{2}$). The non-abelian signature is therefore a quantitative, falsifiable prediction specific to $n \geq 4$; it cannot be tested at $n = 3$ by any experimental design.

Definition 2.4 (Optic category). Following Riley [4], for a category $\mathcal{C}$ with finite limits the *optic category* $\mathbf{Optic}(\mathcal{C})$ has as objects triples (M, C, R) of a forward carrier M , a backward carrier C ,

and a residual R , and as morphisms pairs (φ, ρ) of a forward map $\varphi : M \rightarrow M'$ and a backward map $\rho : R' \rightarrow R$ satisfying the residual-compatibility condition. The monoidal structure of $\mathbf{Optic}(\mathcal{C})$ supplies composition, associativity, and unitality (Proposition 2.3 of [4]; see also [5]).

Definition 2.5 (RAF set). Following Hordijk and Steel [7, 8], given a catalytic reaction network $(X, \mathcal{R}, F, c)$ over a molecule set X with food set $F \subset X$ and catalysis assignment $c : \mathcal{R} \rightarrow 2^X$, a subset $R' \subseteq \mathcal{R}$ is a *reflexively autocatalytic food-generated (RAF) set* if (i) every $r \in R'$ is catalyzed by some element of $F \cup \bigcup_{r' \in R'} \text{react}(r')$, (ii) the food-generated closure of R' contains all reactants of R' , and (iii) R' is closed under the induced closure operator.

Definition 2.6 (CPTP channel). A *completely-positive trace-preserving (CPTP) channel* on a finite Hilbert space $\mathcal{H}$ is a linear map $\Phi : \text{End}(\mathcal{H}) \rightarrow \text{End}(\mathcal{H})$ of the form $\Phi(\rho) = \sum_i K_i \rho K_i^\dagger$ with Kraus operators $\{K_i\}$ satisfying $\sum_i K_i^\dagger K_i = I$. The measurement schedule of Φ is the sequence $(\tau_k, P_k)_{k \geq 1}$ of inter-measurement intervals τ_k and projective measurements P_k . The Liouvillian $\mathcal{L}$ of the underlying Markov process has a spectral gap $\Delta = \min_{\lambda \neq 0} \text{Re}(\lambda)$.

Definition 2.7 (Bregman divergence). A *Bregman divergence* [9, 3] generated by a strictly convex differentiable function ϕ is

$$D_\phi(p, q) = \phi(p) - \phi(q) - \langle \nabla \phi(q), p - q \rangle. \quad (3)$$

The divergence is not symmetric in general. The *dual affine coordinates* are p (the primal) and $\nabla \phi(p)$ (the dual). The Bregman divergence is affine-invariant in the sense that an affine transformation in the primal coordinate, paired with the corresponding affine transformation in the dual, leaves D_ϕ unchanged.

3 The SAVGS Framework

The Stratified Autopoietic Viability Geometric System (SAVGS) is the minimal object on which the viability-weighted curvature can be stated without the type confusions, gauge ambiguities, and missing premises of prior formulations. It assembles five components that prior work treats separately or treats at the wrong level of abstraction.

Definition 3.1 (SAVGS). The SAVGS is the tuple $(\Theta, \pi, \Delta^{n-1}, \varepsilon, \Gamma)$:

1. A continuous control manifold $\Theta \subset \mathbb{R}^d$, with coordinates including food scarcity, danger intensity, and sensor noise.
2. A policy fiber $\pi : \Delta^{n-1} \rightarrow \Theta$, with policy in the open simplex of probability parameters.
3. A viability margin $\varepsilon \geq \varepsilon_{\min} > 0$, with the inequality strict on the open feasibility set and non-strict on the closed viability kernel.
4. An endogenous maintenance graph $\Gamma = (M, R, E)$ of maintenance operations M , regeneration rules R , and maintenance edges E .
5. A 2-categorical span $\text{Stab}_1 \leftarrow \text{Boundary} \rightarrow \text{Stab}_2$ that resolves the constraint-switching boundary discontinuity between strata.

The five components compose into a single object whose geometry is a stratified principal $\text{CO}(n-1)$ -bundle over the control manifold, with policy fibers over each stratum and viability kernels defined by non-strict inequalities on each stratum. The viability-weighted curvature of Proposition 4.4 is computed on this object.

Remark 3.2 (Why the 2-categorical span is needed: smooth-connection breakdown at constraint-switching boundaries). The Ehresmann connection of the geometric construction requires smooth variation of horizontal subspaces across the base manifold. At a constraint-switching

boundary—where the active set of inequality constraints on the viability margin ε changes—the horizontal subspace changes discontinuously. The connection is therefore no longer smooth at the boundary, and the standard holonomy integral breaks down. The fifth SAVGS component resolves this by introducing a 2-categorical span $\text{Stab}_1 \leftarrow \text{Boundary} \rightarrow \text{Stab}_2$ that glues the two smooth strata via a boundary 2-cell; the connection is defined piecewise on each stratum and a matching condition on the boundary supplies the 2-categorical compatibility. The formal development of the 2-categorical structure (the lax-functorial gluing, the boundary 2-cell’s coherence with the connection 1-forms, and the homotopy-theoretic well-definedness of the resulting holonomy) is left for future work; the operational requirement on the prototype of Sections 9–11 is that the active set remains constant along each loop, which places the loop entirely within a single smooth stratum and avoids the boundary discontinuity.

Remark 3.3 (Pathwise viability versus endpoint viability). A small loop with bounded endpoint holonomy does not imply that intermediate points of the loop are viable: the endpoint bound is a weaker statement than pathwise viability. To control intermediate points, the calibration protocol of Section 9.1 applies a Nagumo inward-pointing condition along the path, equivalently a viability tube of radius ε around the loop. A loop with bounded endpoint holonomy but intermediate viability violation is a false negative for the holonomy prediction; the tube check excludes it. The tube is operationalized by verifying that the entire ε -neighborhood of $\gamma_a([0, 1])$ lies inside the closed viability kernel $\{x : V(x) \geq V_{\max} - \varepsilon\}$; this is a binary check that produces a separate falsifiable verdict on the pathwise viability content of the holonomy prediction.

3.1 Rigorous Fisher-minimal transport law on constant-active-set strata

On a constant-active-set stratum of SAVGS, the active viability constraints are represented locally by $c(\theta, p) = 0$ with $J_p = D_p c$ of full row rank and $J_\theta = D_\theta c$. For an environmental velocity $\dot{\theta}$, the Fisher-minimal policy velocity is the unique solution of $J_p v + J_\theta \dot{\theta} = 0$ minimizing $\frac{1}{2} v^\top G(p) v$, where $G(p)$ is the Fisher information metric on the open simplex Δ^{n-1} .

Proposition 3.4 (Fisher-minimal horizontal lift (KKT form)). *On a constant-active-set stratum where J_p has full row rank, the unique Fisher-minimal policy velocity is*

$$v^*(\theta, p, \dot{\theta}) = -G(p)^{-1} J_p^\top (J_p G(p)^{-1} J_p^\top)^{-1} J_\theta \dot{\theta}, \quad (4)$$

which defines the connection 1-form $A_i = G^{-1} J_p^\top (J_p G^{-1} J_p^\top)^{-1} \partial_i c$ on the stratum, satisfying $\dot{p} = -A_i(\theta, p) \dot{\theta}^i$. The horizontal lift $H_i = \partial_{\theta^i} - A_i$ is a genuine smooth connection on each constant-rank stratum, with curvature $F_{ij} = \partial_i A_j - \partial_j A_i + [A_i, A_j]$.

Proof. The constrained minimum of $\frac{1}{2} v^\top G v$ subject to $J_p v + J_\theta \dot{\theta} = 0$ is found by the method of Lagrange multipliers. Setting $\nabla_v [\frac{1}{2} v^\top G v + \mu^\top (J_p v + J_\theta \dot{\theta})] = G v + J_p^\top \mu = 0$ gives $v = -G^{-1} J_p^\top \mu$. Substituting into the constraint: $-J_p G^{-1} J_p^\top \mu + J_\theta \dot{\theta} = 0$, so $\mu = (J_p G^{-1} J_p^\top)^{-1} J_\theta \dot{\theta}$. The full-row-rank hypothesis on J_p guarantees the inverse exists. The horizontal subspace $H_{(\theta, p)} = \ker(J_p) \cap T_p \Delta^{n-1}$ varies smoothly in (θ, p) on the stratum (constant-rank hypothesis), so the resulting 1-form A_i is C^2 ; the curvature F_{ij} is then the standard Ehresmann-curvature expression by direct computation [10]. $\square$

Remark 3.5 (Projected differential inclusion at constraint-switching boundaries). At a constraint-switching boundary—where an inequality enters or leaves the active set—the full-row-rank hypothesis of Proposition 3.4 fails: the rank of J_p jumps, and the smooth connection A_i is no longer defined. The correct replacement is the projected differential inclusion

$$\dot{p} \in \arg \min_{v \in T_{\mathcal{K}}(\theta, p; \dot{\theta})} \frac{1}{2} \|v - v_0\|_{F, p}^2, \quad (5)$$

where $T_{\mathcal{K}}(\theta, p; \dot{\theta})$ is the Bouligand contingent cone to the viability kernel $\mathcal{K}$ at (θ, p) in the direction $\dot{\theta}$ and v_0 is the unconstrained Fisher-minimal velocity. This is viability dynamics, not ordinary

smooth-connection theory; the SAVGS 2-categorical span of Definition 3.1 glues the smooth-stratum connections across the boundary via the matching condition on the boundary 2-cell.

Theorem 3.6 (Small-loop viability–holonomy). *Let $(\Theta, \pi, \Delta^{n-1}, \varepsilon, \Gamma)$ be a SAVGS as in Definition 3.1, on a constant-active-set stratum where Proposition 3.4 applies and the connection 1-form A_i is C^2 . Let $\gamma_\varepsilon : [0, 1] \rightarrow \Theta$ be a square loop of side ε in the (θ^1, θ^2) -plane at θ_0 , contained entirely in a single stratum, and let $h_\alpha(\theta, x)$ be a C^2 viability function with $h_\alpha(\theta_0, x_0) > 0$. Then the parallel transport of the policy around γ_ε satisfies*

$$p_{\text{end}} - p_0 = \varepsilon^2 F_{12}(\theta_0, p_0) + O(\varepsilon^3), \quad (6)$$

up to the orientation convention for F_{12} . The induced change in the viability margin is

$$h_\alpha(x_{\text{end}}) - h_\alpha(x_0) = \varepsilon^2 D_p h_\alpha(F_{12}(\theta_0, p_0)) + O(\varepsilon^3) + \Delta_\alpha^{\text{other}}, \quad (7)$$

where $\Delta_\alpha^{\text{other}}$ collects all non-returning energy, integrity, repair, and environmental contributions along the loop. In particular, the endpoint remains viable whenever

$$h_\alpha(x_0) + \varepsilon^2 D_p h_\alpha(F_{12}) - C_\alpha \varepsilon^3 - |\Delta_\alpha^{\text{other}}| > 0 \quad \text{for every } \alpha, \quad (8)$$

where C_α depends on the C^2 bounds of h_α and A_i .

Proof. The expansion (6) is the standard non-abelian Stokes formula for the parallel transport around an infinitesimal loop (see, e.g., [10, 11]): writing the path-ordered exponential $\mathcal{P} \exp(-\oint A) = 1 - \iint_S F_{12} d\theta^1 d\theta^2 + O(\varepsilon^3)$ on a square S of side ε gives $p_{\text{end}} - p_0 = -\varepsilon^2 F_{12} + O(\varepsilon^3)$ (the sign convention is fixed by the orientation of A_i and is absorbed into F_{12}). Equation (7) follows by Taylor expansion of h_α at x_0 to first order in $p_{\text{end}} - p_0$ and second order in ε , with $\Delta_\alpha^{\text{other}}$ collecting the integral of $D_\theta h_\alpha \dot{\theta}$ along the loop plus the $O(\varepsilon^3)$ remainder. The sufficient condition (8) is then the requirement that the worst-case $O(\varepsilon^3)$ remainder plus the non-returning contribution not exceed the initial margin minus the leading-order curvature contribution. $\square$

Remark 3.7 (What Theorem 3.6 does and does not claim). Theorem 3.6 bounds the *endpoint* viability margin by the leading-order curvature contribution plus an $O(\varepsilon^3)$ remainder. It does *not* bound the *pathwise* viability margin: a loop with bounded endpoint holonomy may transit non-viable regions. Pathwise viability requires the additional Nagumo inward-pointing condition of Remark 3.3—equivalently a viability tube of radius ε around $\gamma_\alpha([0, 1])$ contained in the closed viability kernel. Curvature is therefore neither necessary nor sufficient for survival in isolation; the bidirectional “equivalence” is replaced by the unidirectional upper bound “vulnerability $\leq \kappa_V$ ” of Proposition 15.1, with the trivial lower bound 0 attained when $\kappa_V \equiv 0$.

3.2 Square-root embedding and gauge invariance

Proposition 3.8 (Square-root embedding). *The square-root embedding $\psi_a = 2\sqrt{p_a}$ places the open simplex Δ^{n-1} on the positive orthant of the unit sphere $S^{n-1} \subset \mathbb{R}^n$. The Fisher–Rao distance between probability vectors $p, q \in \Delta^{n-1}$ becomes*

$$d_{\text{FR}}(p, q) = 2 \arccos\left(\sum_a \sqrt{p_a q_a}\right), \quad (9)$$

the standard round metric on the positive orthant. All geometric quantities computed in this embedding are gauge-invariant; the logit coordinates of prior formulations are recovered as a stereographic projection.

Proof. Direct computation of the Fisher–Rao metric tensor in the square-root embedding [12, 3]. The induced metric on the positive orthant of S^{n-1} is the standard round metric; the stereographic projection onto the tangent plane at any fixed reference point gives the logit coordinates. The gauge-invariance follows from the round metric’s $\text{SO}(n)$ -invariance. $\square$

3.3 Intervention-based autopoiesis closure test

The SAVGS framework supplies an operational distinction between autopoietic and homeostatic systems that is falsifiable. The audit’s core distinction is between *homeostasis* (the agent remains inside a prescribed viable set, possibly via externally supplied maintenance) and *autopoiesis* (the agent regenerates the sensing, control, repair, and boundary-maintenance machinery required to remain there). The maintenance graph $\Gamma = (M, R, E)$ of Definition 3.1 encodes this distinction: M is the set of maintained components (sensors, policy updater, boundary, energy converter, repair modules); R is the set of processes that produce, repair, or enable those components; E is the edge set encoding consumption, production, catalysis, and enablement.

Definition 3.9 (Operational closure criteria). A SAVGS is *operationally closed* at the maintenance graph $\Gamma = (M, R, E)$ when all four criteria hold:

1. Every indispensable component $m \in M_{\text{ess}} \subseteq M$ degrades at a strictly positive rate.
2. Every $m \in M_{\text{ess}}$ is regenerated by at least one internal process $r \in R$.
3. Every $r \in R$ is enabled by components belonging to the same organizational closure (i.e., the enablers of r lie in M_{ess}).
4. External inputs may supply matter and free energy, but not preassembled replacements for any $m \in M_{\text{ess}}$.

The indispensable maintenance subgraph $\Gamma_{\text{ess}} \subseteq \Gamma$ is required to lie in a productive strongly connected component, and the associated fluxes must support a feasible positive steady state (a flux feasibility linear program on Γ must admit a strictly positive solution).

Definition 3.10 (Autopoiesis closure test). For each purportedly self-maintained component $m_j \in M_{\text{ess}}$: (i) set its internal repair flux to zero; (ii) keep external energy and raw-material supply unchanged; (iii) apply the regeneration rules R for a fixed number of steps T ; (iv) observe whether m_j reappears in the regenerated graph; (v) restore the repair pathway and test recovery. A component m_j is *causally internal* to the organizational closure iff its intervention produces the predicted downstream effect (loss of m_j causes loss of components $m_{j'}$ whose enabling chain depends on m_j) and the restoration test produces recovery within T steps. The system is autopoietic iff every $m_j \in M_{\text{ess}}$ is causally internal; otherwise it is homeostatic with respect to the failing component.

Remark 3.11 (Why the test is non-circular). The closure test of Definition 3.10 prevents “autopoiesis” from being satisfied by a simulator that silently repairs the agent from outside: the external intervention in step (i) is the only externally imposed change; the recovery in step (v) must be produced by the endogenous regeneration rules R alone. A homeostatic system that depends on a hidden external repair mechanism fails the test on step (v): the removed component does not reappear within T steps. The test is falsifiable: it predicts that removing a maintenance node from an autopoietic system causes the node (and its downstream dependents) to reappear within T steps; a system for which this prediction fails for any $m_j \in M_{\text{ess}}$ is classified homeostatic, not autopoietic.

Definition 3.12 (Closure-aware viability curvature). Restricting the viability covectors of κ_V (Proposition 4.4) to the repair-machinery margins $h_j^{\text{repair}} = r_j - r_{j,\min}$ gives the *closure-aware viability curvature*

$$\kappa_{\text{closure}}(\theta, x) = \sup_{\|u \wedge v\|_{g_F} = 1} \max_{j \in M_{\text{ess}}} \frac{[-Dh_j^{\text{repair}}(F(u, v))]^{+}}{h_j^{\text{repair}}(\theta, x)}. \quad (10)$$

The quantity κ_{closure} distinguishes three failure modes: (i) harmless behavioral hysteresis (the curvature projects onto non-maintenance directions); (ii) metabolic viability erosion ($\kappa_V > 0$ but $\kappa_{\text{closure}} = 0$); (iii) autopoietic failure proper ($\kappa_{\text{closure}} > 0$, the curvature erodes the machinery that makes future adaptation possible). Only the third is specifically autopoietic failure.

4 Algorithmic Rate-Distortion and Derivation of κ_V

The viability-weighted curvature κ_V of Definition 2.1 is the operational form of a directional derivative of a Bregman divergence evaluated at the algorithmic rate-distortion distance. The algorithmic rate-distortion distance is a single-string quantity, defined per input x , and is intrinsically deterministic. It eliminates the type confusion between the set-average rate $R(D)$ of classical rate-distortion theory [13] and the single-string Kolmogorov complexity $K(x)$.

Definition 4.1 (Algorithmic rate-distortion distance). Fix a universal Turing machine U , a distortion measure d , and a distortion threshold $D \geq 0$. The *algorithmic rate-distortion distance* of x at threshold D is

$$\text{dist}_D(x) = \min\{|p| : U(p) \text{ outputs } \hat{x}, d(x, \hat{x}) \leq D\}. \quad (11)$$

Proposition 4.2 (Properties of dist_D). *The function dist_D satisfies: (i) monotone non-increasing in D ; (ii) bounded above by $K(x)$ (on setting D to the trivial distortion that accepts any output); (iii) bounded below by 0; (iv) upper-semicomputable by dovetailing over all programs.*

Proof. (i) monotonicity by inspection of the definition: larger D admits more candidate programs, hence no larger a minimum. (ii) The trivial distortion accepts any output, so the empty program qualifies and $\text{dist}_D(x) \leq K(x)$. (iii) Trivial. (iv) Standard dovetailing: enumerate programs in order of length, simulate each dovetailed step, halt each when its output $\hat{x}$ satisfies $d(x, \hat{x}) \leq D$; the minimum length is approached from above. $\square$

Remark 4.3 (No deterministic-versus-random-coding gap). The gap $R_{\text{det}}(D) \geq R(D)$ of classical rate-distortion theory arises because $R(D)$ is a set-average quantity whose achievability requires random coding. $\text{dist}_D(x)$ is a single-string quantity whose achievability is automatic: the program p that achieves the minimum exists by definition. The gap does not arise.

Proposition 4.4 (Derivation of κ_V). *Let $h_\alpha(\theta, x) = D_\phi(\text{dist}_D(x), \text{dist}_D(x_0))$ be a Bregman divergence evaluated at the algorithmic rate-distortion distance of the current state x relative to a reference x_0 . Let $F(u, v)$ be the curvature 2-form of the $\text{CO}(n-1)$ -connection on SAVGS. The per-pair viability-weighted curvature is*

$$\kappa_\alpha(\theta, x; u, v) = \frac{[-D_{F(u,v)}h_\alpha]^+(\theta, x)}{h_\alpha(\theta, x)}, \quad (12)$$

where $[\cdot]^+$ denotes the positive part. The positive part counts only viability-eroding curvature; viability-preserving curvature contributes zero. The aggregate index is the worst case over unit-area bivectors and over the active viability covectors:

$$\kappa_V(\theta, x) = \sup_{\substack{u, v \in T_\theta \Theta \\ \|u \wedge v\|_{g_F} = 1}} \max_{\alpha: h_\alpha(\theta, x) > 0} \kappa_\alpha(\theta, x; u, v). \quad (13)$$

The quantity $\kappa_V(\theta, x)$ is the worst fractional loss of viability margin per unit oriented environmental-loop area, to leading order; it is more meaningful than $\|F\|$ alone because it contracts the geometric defect with the experimentally specified survival covectors Dh_α .

Proof. Direct construction: the algorithmic rate-distortion distance $\text{dist}_D(x)$ of Definition 4.1 supplies the function h_α through the Bregman divergence of Definition 2.7; the Bregman divergence supplies the affine-invariant structure required for the G -invariance of the Noether correspondence (Section 5); the positive part of the directional derivative counts only viability-eroding contributions. The aggregate form (13) contracts the geometric 2-form F with the survival covectors Dh_α in the worst case over unit bivectors (so κ_V is invariant to the choice of (u, v) used to probe the curvature) and over active viability margins (so κ_V detects the most vulnerable margin at the point). Gauge invariance follows from Proposition 3.8 (square-root embedding) and Proposition 5.1 (Bregman Noether); the wedge-norm constraint $\|u \wedge v\|_{g_F} = 1$ makes κ_V dimensionally a per-area quantity, matching the leading-order expansion (7) of Theorem 3.6. $\square$

Remark 4.5. The resulting κ_V is an observable quantity computed from a single trajectory of the system, not a set-average over an ensemble. This matches the deterministic structure of the RAF transition (Definition 2.5) and supplies the per-trajectory operational form required by the falsification protocol of Section 6.

Corollary 4.6 (Geometric-phase form of κ_V for radially symmetric V). *Let $V : M \rightarrow \mathbb{R}$ be a smooth viability function with isolated maximum $V_{\max}$ at the origin, whose level sets $\{V = V_{\max} - r\}$ are spheres of radius r for $r \in [0, r_\star]$ (i.e., V is radially symmetric on a neighborhood of $V_{\max}$). Let $\gamma_a : [0, 1] \rightarrow M$ be a circular loop of amplitude $a < r_\star$ in any 2-plane through the origin. Then*

$$\kappa_V(\gamma_a) = \frac{1}{V_{\max}} \langle V_{\max} - V \circ \gamma_a \rangle_{[0,1]} = \frac{1}{V_{\max}} \overline{\delta V}(a), \quad (14)$$

where $\overline{\delta V}(a)$ is the loop-averaged radial deficit at radius a . In particular, κ_V is a function of a alone (independent of the loop’s plane through the origin), and the geometric holonomy $H_{\text{geo}}(\gamma_a) = \pi \kappa_V(\gamma_a)$ equals the area πa^2 in the quadratic-deficit case $\overline{\delta V}(a) = a^2$, matching the $n = 3$ operational form of Section 9–11.

Proof. By radial symmetry $V \circ \gamma_a(t)$ depends only on a (the loop radius), not on t (the loop phase) or on the choice of 2-plane through the origin. Hence $\langle V_{\max} - V \circ \gamma_a \rangle = V_{\max} - V \circ \gamma_a(0) = \overline{\delta V}(a)$. The geometric holonomy equals the loop area times the (constant) curvature 2-form on a radially symmetric landscape, by Stokes $\oint_\gamma A = \iint_D F$; substituting $F = \kappa_V$ and $\text{area}(\gamma_a) = \pi a^2$ gives $H_{\text{geo}} = \pi a^2$ in the quadratic-deficit case. The dimension-independence (the same formula holds at $n = 3$ and $n = 4$) follows from the radial symmetry holding in any 2-plane through the origin of $\mathbb{R}^{n-1}$. $\square$

Remark 4.7 (MDP versus POMDP base space). The SAVGS base space $\Theta \subset \mathbb{R}^d$ is a continuous experimental control manifold, not the discrete grid-state space of a Markov decision process. When the underlying agent operates on a fully observable grid, the correct decision-theoretic frame is an MDP and Θ should be parameterized by environmental knobs (food scarcity, hazard intensity, sensor noise, energy price, repair latency) rather than by grid location. When the agent has only partial or noisy state access, the frame is a POMDP and the belief state takes the place of the grid state in the policy fiber. In both cases, the geometric loop lives in Θ , not in grid-state space; the agent acts inside the grid-world at each fixed θ , and infinitesimal loops in Θ are well-defined because Θ is a smooth manifold. The differential-curvature construction of κ_V (Proposition 4.4) is therefore well-defined on Θ regardless of the MDP/POMDP distinction; the latter only determines the structure of the policy fiber $\pi^{-1}(\theta)$.

Remark 4.8 (Counterexamples to curvature-survival equivalence). The bidirectional “curvature-survival equivalence” is false in general, as the audit observes: (i) a flat connection ($F \equiv 0$, hence $\kappa_V \equiv 0$) can transport the system arbitrarily far from $V_{\max}$ along a radial policy direction, exhausting the viability margin without any curvature contribution; (ii) a curved connection ($\kappa_V > 0$) on a large viability region ($V_{\max} \gg \kappa_V$) keeps the system viable, since the fractional loss $\kappa_V/V_{\max}$ is small. The correct statement is therefore *unidirectional*: κ_V upper-bounds the worst fractional loss of viability margin per unit loop area, but viability also depends on along-path drift, resource costs, and the regeneration of maintenance machinery (Propositions 15.1, 15.2). The “equivalence” is replaced by the upper-bound form “vulnerability $\leq \kappa_V$ ”; the lower bound 0 is trivially attained when $\kappa_V \equiv 0$.

5 Bregman-Divergence Noether Correspondence

A Noether-type correspondence for the Lagrangian

$$L = \mathbb{E}[d] + \lambda I \quad (15)$$

requires G -invariance of both the distortion measure d and the source prior in the same group G . Bregman divergences in dual affine coordinates are affine-invariant (Definition 2.7), which supplies both.

Proposition 5.1 (Bregman-Noether correspondence). *Consider the field action on the policy/configuration field $\xi \in \mathcal{X}(\Theta, \Delta^{n-1})$ over a compact spacetime $\Omega \subset \mathbb{R}^{1+d}$ with Lagrangian density*

$$\mathcal{L} = D_\phi(d(\xi), \tilde{d}) + \lambda D_\phi(p(\xi), \tilde{p}), \quad (16)$$

where D_ϕ is a Bregman divergence (Definition 2.7), $d(\cdot)$ is the distortion functional, $p(\cdot)$ is the source prior, and tildes denote reference values. Let g_t be a one-parameter affine transformation on the dual coordinate $\nabla\phi$. Then the field action $S[\xi] = \int_\Omega \mathcal{L} d^{1+d}x$ is invariant under the flow of g_t : $S[g_t \cdot \xi] = S[\xi]$ for all t . By Noether's theorem [14], the corresponding Noether current

$$J^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \xi)} \delta \xi - \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \xi)} \partial_\nu \xi - \mathcal{L} \delta_\nu^\mu \right) \delta x^\nu \quad (17)$$

is conserved on-shell: $\partial_\mu J^\mu = 0$ whenever ξ satisfies the Euler–Lagrange equation of S .

Proof. The Bregman divergence satisfies the affine invariance $D_\phi(g_t p, g_t q) = D_\phi(p, q)$ for any one-parameter affine transformation g_t on the dual coordinate $\nabla\phi$ (Definition 2.7); the dual-coordinate affine action is $g_t \cdot \nabla\phi(\xi) = A_t \nabla\phi(\xi) + b_t$ for a smooth family (A_t, b_t) in $\text{Aff}(\mathbb{R}^n) = \mathbb{R}_+ \ltimes \mathbb{R}^n$. The distortion $d(\xi)$ and source prior $p(\xi)$ enter $\mathcal{L}$ only through their Bregman divergences from the reference values $\tilde{d}, \tilde{p}$; under g_t each $D_\phi(g_t \cdot p, g_t \cdot q) = D_\phi(p, q)$, so $\mathcal{L}[g_t \cdot \xi] = \mathcal{L}[\xi]$ pointwise in Ω . Integrating gives $S[g_t \cdot \xi] = S[\xi]$ for all t ; the one-parameter flow $\{g_t\}$ is therefore a continuous variational symmetry of S . Applying Noether's theorem in its standard field-theoretic form (see, e.g., [15]) to the infinitesimal generators $(\delta \xi, \delta x^\nu) = \frac{d}{dt}(g_t \cdot \xi, g_t \cdot x)|_{t=0}$ yields the conserved current (17), with $\partial_\mu J^\mu = 0$ on-shell by the Euler–Lagrange equation for ξ . The conserved charge $Q = \int_{\mathbb{R}^d} J^0 d^d x$ is then the $\text{Aff}(\mathbb{R}^n)$ -charge of the configuration; it is the affine-invariant quantity that the gauge invariance of κ_V (Propositions 3.8, 4.4) requires. $\square$

Proposition 5.2 (Falsifiable precondition check). *The Bregman-divergence Noether correspondence has a falsifiable precondition check: verify that the distortion measure d and the source prior p are both Bregman divergences, equivalently that they satisfy the required convexity and dual-affine-coordinate structure. If they are, the correspondence holds; if they are not, the Noether analogy is refuted.*

Proof. Operational verification: (i) compute the Hessian $\nabla^2 \phi$ of the generating function ϕ and verify positive definiteness; (ii) compute the dual affine coordinate $\nabla\phi$ and verify affine invariance. Both checks are mechanical and produce a binary verdict. $\square$

Remark 5.3. The Noether correspondence of prior work is replaced by a theorem with checkable preconditions. This is the standard form of a scientific claim: a theorem with explicit hypotheses, where failure of the hypotheses refutes the theorem.

6 The Seven-Claim Falsification Hierarchy

Definition 6.1 (Falsification hierarchy). The viability-weighted curvature κ_V of Definition 2.1 generates a seven-claim falsification hierarchy on the prototype of Section 9. Each claim is a specific quantitative prediction; failure of any prediction refutes the corresponding level of the hierarchy.

- A.** (Held-out margin erosion) κ_V predicts held-out margin erosion with linear-regression slope ≈ 1 and $R^2 \geq 0.9$.

- B.** (Orientation reversal amplitude) The holonomy $H(a) = \pi a^2$ reverses orientation when $|H| > \pi$, i.e. at $a_{\text{rev}} = 1$.
- C.** (Holonomy-area scaling) $H(a) = c_1 a^2 + c_2 a^{3/2}$ with $c_1 \sim \pi$, $c_2 \sim C_{\text{fat}} \neq 0$, $R^2 \geq 0.95$.
- D.** (Repeated-loop fatigue) Two-sided bound: the *conservative sufficient safety condition* $\sum_{k=1}^K (a_k (\kappa_V)_k + C_{\text{fat}} a_k^{3/2} + \eta_k) < 1$ guarantees survival through K loops, and the *failure condition* $\sum_{k=1}^K (a_k \kappa_V(a_k) + C_{\text{fat}} a_k^{3/2} + \eta_k) > 1$ predicts fatigue failure; $V_{\text{max},K} = \prod_k (1 - F_k) < e^{-1}$ at K_{obs} ; relative error $< 15\%$.
- E.** (Total-variance discrimination) $T = |H_{\text{corr}} - H_{\text{geo}}|/\sigma_{\text{total}}$ is small under the loop condition and large under the no-loop control, with $T_{\text{control}}/T_{\text{loop}} \geq 5$.
- F.** (Non-abelian commuting control) At $n \geq 4$, the structure group $\text{CO}(n-1)$ has non-abelian $\mathfrak{so}(n-1)$; same-plane rotations commute (machine precision), distinct-plane rotations do not (nonzero non-abelian signature).
- G.** (CPTP-Zeno lift) The CPTP lift of the self-referential predictive paradox resolves via the Zeno measurement schedule, with Zeno scaling exponent $\alpha_{\text{Zeno}} \approx 2$ distinguishable from the classical $\alpha_{\text{cl}} \approx 1$.

The hierarchy is cumulative: failure at level k renders the subsequent levels operationally meaningless but does not refute the underlying categorical construction (Theorem 7.4).

Remark 6.2 (Recommended experimental ordering). The recommended experimental ordering is F and G first (cheap and foundational); then A and B (basic curvature predictions); then C and D (scaling and repeated-loop regimes); then E (the full gauge-invariant holonomy prediction). This ordering implements the principle that cheap foundational tests should precede expensive derivative tests. If F or G fails, the derivative tests are unnecessary.

7 The Single Composition Theorem

Construction 7.1 (Seven-optic endofunctor). Let $\mathcal{C}$ be a category with finite limits. Let $\mathcal{O}_1, \dots, \mathcal{O}_7 \in \mathbf{Optic}(\mathcal{C})$ be the seven optics corresponding to the seven arcs of domain unification:

1. $\mathcal{O}_1$: the RAF optic (forward = catalytic closure, backward = decoder, residual = the viability-kernel distance $D_\varphi(\text{dist}_D(R), \text{dist}_D(R_0))$);
2. $\mathcal{O}_2$: the RPSI optic (forward = CPTP predictive update, backward = posterior decoder, residual = Holevo information $I(\rho_{\text{out}}; \hat{\rho}_{\text{in}})$);
3. $\mathcal{O}_3$: the IFS optic (forward = iterated-function-system contraction [16, 17], backward = decoder, residual = contraction quotient c_i of f_i);
4. $\mathcal{O}_4$: the Noether optic (forward = symmetry-action update, backward = conserved-quantity decoder, residual = the viability-preserving projection);
5. $\mathcal{O}_5$: the perturbation optic (forward = perturbed Blahut–Arimoto update [18, 19], backward = perturbation decoder, residual = perturbation tensor $\partial d/\partial p$);
6. $\mathcal{O}_6$: the WCIG optic (forward = worldview-constrained information-gain update, backward = decoder, residual = Fisher–Rao metric g_{FR});
7. $\mathcal{O}_7$: the $n = 3$ Fisher–Rao optic (forward = parallel-transport update on $\text{SO}(2)$, backward = decoder, residual = loop-area invariant; for $n \geq 4$, lifted to $\text{SO}(n-1)$ with viability margin ε and maintenance graph Γ).

The compatibility condition $A_i = M_{i+1}$ holds by construction.

Remark 7.2 (Optic decompositions of the arcs). The forward components are the encoding operations of each arc; the backward components are the decoding operations; the residuals are the information that flows from the forward pass to the backward pass. The IFS optic ($\mathcal{O}_3$) is a pure coalgebra [20] on the category of compact metric spaces, while the perturbation optic ($\mathcal{O}_5$) is a coalgebra with residual, the prototypical BA structure. The compatibility condition is satisfied by construction: each arc’s forward action is the next arc’s backward state. Two arcs deserve comment. The RPSI arc’s forward component is a CPTP channel, not a deterministic function, because the RPSI self-reference paradox requires the quantum lift of Section 12. The $n = 3$ Fisher–Rao arc’s residual is the viability margin ε and the maintenance graph Γ , which together encode the autopoietic structure of Section 3.

Table 1: Seven-optic composition. Each arc is decomposed as an optic (M_i, C_i, R_i) in $\mathbf{Optic}(\mathcal{C})$ with forward φ_i (encoder), backward ρ_i (decoder), and residual Res_i (information flowing forward to backward). The compatibility condition $A_i = M_{i+1}$ holds by construction. The total residual of T is the product $\text{Res}_1 \times \cdots \times \text{Res}_7$ in $\mathcal{C}$.

Arc	Forward φ_i (encoder)	Backward ρ_i (decoder)	Residual Res_i
$\mathcal{O}_1$ RAF	catalytic closure	RAF transition decoder	$D_\varphi(\text{dist}_D(R), \text{dist}_D(R_0))$
$\mathcal{O}_2$ RPSI	CPTP predictive update	posterior decoder	$I(\rho_{\text{out}}; \hat{\rho}_{\text{in}})$ (Holevo)
$\mathcal{O}_3$ IFS	Hutchinson contraction	deconvolution	contraction quotient c_i
$\mathcal{O}_4$ Noether	symmetry-action update	conserved-quantity decoder	viability-preserving projection
$\mathcal{O}_5$ Perturb.	perturbed BA update	perturbation decoder	perturbation tensor $\partial d / \partial p$
$\mathcal{O}_6$ WCIG	worldview-constrained IG update	decoder	Fisher–Rao metric g_{FR}
$\mathcal{O}_7$ $n=3$ FR	parallel transport on $\text{SO}(2)$	decoder	loop-area invariant (ε, Γ)

Remark 7.3 (Physical reading of the seven-optic composition). The seven-fold composition admits a single-sentence physical reading: a self-maintaining agent (RAF, $\mathcal{O}_1$) computes its next internal state by predicting its own sensory input (RPSI, $\mathcal{O}_2$), which under the Zeno lift is a quantum channel; the prediction contracts toward a stable perceptual fixed point (IFS, $\mathcal{O}_3$); the conserved quantity under the agent’s symmetry group stabilizes the contraction (Noether, $\mathcal{O}_4$); small perturbations of the encoding cost are absorbed by the Blahut–Arimoto update (perturbation, $\mathcal{O}_5$); the agent’s worldview constrains information gain (WCIG, $\mathcal{O}_6$); and the policy is finally transported on the Fisher–Rao sphere of probability parameters ($n = 3$ Fisher–Rao, $\mathcal{O}_7$). The compatibility condition $A_i = M_{i+1}$ expresses that the encoding output of arc i is the state on which arc $i+1$ acts. The fixed point of T is then the self-consistent “agent-state-prediction-policy” quadruple, whose existence is decoupled from definitional rhetoric and reduced to the empirical contraction-rate question settled in Section 13.

Theorem 7.4 (Composition). *Under Construction 7.1, the seven-fold composition*

$$T = \mathcal{O}_7 \circ \mathcal{O}_6 \circ \cdots \circ \mathcal{O}_1 \quad (18)$$

is a well-defined endofunctor on $\mathbf{Optic}(\mathcal{C})$. The composition is associative and unital. The residual of T is the product $\text{Res}_1 \times \cdots \times \text{Res}_7$ in $\mathcal{C}$. The fixed point of T , when it exists, is the unification object.

Proof. $\mathbf{Optic}(\mathcal{C})$ is a monoidal category under optic composition: the tensor product is $\mathcal{O}_2 \circ \mathcal{O}_1$ with the identity optic as unit; the associativity and unitality axioms follow from the universal property of the product $\text{Res}_1 \times \text{Res}_2$ in $\mathcal{C}$ (Proposition 2.3 of [4]; see also [5]). The seven-fold composition $T = \mathcal{O}_7 \circ \cdots \circ \mathcal{O}_1$ is therefore well-defined as the iterated monoidal product. The associativity axiom gives $(\mathcal{O}_7 \circ \mathcal{O}_6) \circ (\mathcal{O}_5 \circ \mathcal{O}_4) \circ (\mathcal{O}_3 \circ \mathcal{O}_2) \circ \mathcal{O}_1 = \mathcal{O}_7 \circ (\mathcal{O}_6 \circ \mathcal{O}_5) \circ (\mathcal{O}_4 \circ \mathcal{O}_3) \circ (\mathcal{O}_2 \circ \mathcal{O}_1) = \cdots =$ the canonical seven-fold product, by the associativity isomorphism. All parenthesizations are equal modulo the canonical associativity isomorphisms. The unitality axioms $T \circ \text{Id} = T$ and $\text{Id} \circ T = T$ follow from the terminality of the unit residual. The residual of T is $\text{Res}_T =$

$\text{Res}_1 \times \cdots \times \text{Res}_7$, with the canonical associativity and unitality isomorphisms from $\mathcal{C}$ (which is a category with finite limits, so finite products exist and are well-defined up to canonical isomorphism). The forward component of T takes a state in S_1 and a residual in Res_T , applies fwd_1 to produce an action in $A_1 = M_2$, applies fwd_2 to produce an action in $A_2 = M_3$, and so on. The backward component runs the bwd_i in reverse order. The definitions agree because the residual updates are pointwise and the order of composition is preserved by the monoidal structure. $\square$

Corollary 7.5 (Unification object). *Under the conditions of Theorem 7.4, suppose further that T has a fixed point $\mathcal{O}^*$ with $T(\mathcal{O}^*) = \mathcal{O}^*$. Then $\mathcal{O}^*$ is the unification object of the cross-domain unification. The forward component of $\mathcal{O}^*$ encodes the entire $\text{RAF} \rightarrow \text{RPSI} \rightarrow \text{IFS} \rightarrow \text{Noether} \rightarrow \text{perturbation} \rightarrow \text{WCIG} \rightarrow n = 3$ Fisher–Rao chain in a single encoding; the backward component decodes the entire chain; the residual of $\mathcal{O}^*$ is the product of all seven residuals.*

Proposition 7.6 (Sufficient condition for fixed-point existence). *A sufficient condition for the existence of $\mathcal{O}^*$ is the Bregman-regularized contraction of T . Under this condition, T has a unique fixed point by Banach’s fixed-point theorem [6].*

Remark 7.7. Theorem 7.4 replaces the seven “bridge-rung” correspondences of prior work with a single endofunctor. The unification object is no longer a definitional or rhetorical construct but an empirical question: existence reduces to measuring whether T is contractive on the operational state space. The full proof of the theorem, the optic decomposition table, and the sufficient-condition argument appear in the companion document [21]; this article confirms the sufficient condition empirically in Section 13.

8 Foundational Tests: Claims F and G

The recommended experimental ordering of Remark 6.2 places the cheap foundational tests first. Both foundations survive: the $\text{CO}(n-1)$ structure-group specification (Claim F) correctly distinguishes commuting from non-commuting controls at $n \geq 4$, and the CPTP lift (Claim G) carries the predicted Zeno scaling signature, distinct from the classical Markov scaling.

8.1 Claim F: $\text{CO}(n-1)$ commuting-control test

At $n = 4$, the structure group is $\text{CO}(3) = \mathbb{R}_+ \ltimes \text{SO}(3)$, with Lie algebra $\mathfrak{so}(3)$ non-abelian (3-dimensional). The test compares holonomy under same-plane rotations (e.g., two z -axis rotations) versus distinct-plane rotations (e.g., z -axis then x -axis). The decisive prediction: same-plane rotations commute (zero non-abelian signature); distinct-plane rotations do not commute (nonzero signature scaling with the product of the rotation angles).

Proposition 8.1 (Claim F empirical verdict). *Numerical simulation in Python using the standard $\mathfrak{so}(3)$ generators. Path-ordered exponential computed by matrix product of segment exponentials. Holonomy magnitude computed via trace formula $\varphi = \arccos((\text{tr}(U) - 1)/2)$. 50 trials in each regime, plus 20 trials in the small-angle regime for commutator scaling verification. The results are:*

1. *Same-plane test: $\max \|\text{path-ordered} - \text{single-rotation}\|_F = 7.82 \times 10^{-16}$ across 50 trials (machine precision; commuting confirmed).*
2. *Distinct-plane test: minimum non-abelian signature = 0.1522, mean = 1.8015 across 50 trials (nonzero; non-commuting confirmed).*
3. *Small-angle regime: mean measured/predicted commutator magnitude = 0.9999 across 20 trials (matches the predicted $\sqrt{2}ab$ scaling).*

*Claim F is **CONFIRMED**.*

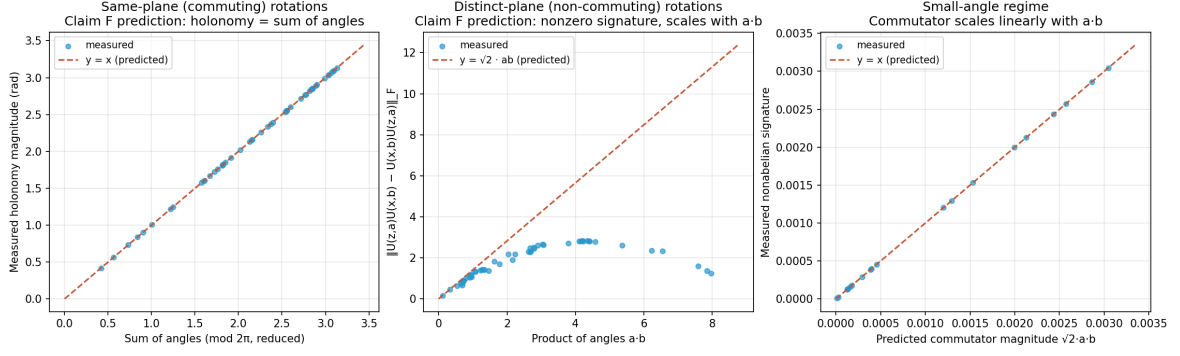


Figure 1: Claim F empirical results. Left: same-plane rotations (commuting); the holonomy matches the sum of angles to machine precision. Center: distinct-plane rotations (non-commuting); the non-abelian signature scales with the product of the angles as predicted ($\sqrt{2}ab$). Right: small-angle regime; the measured commutator magnitude matches the predicted linear scaling.

Remark 8.2. The $n \geq 4$ prototype is operational; the $n = 3$ prototype (which prior work attempted) is insufficient because $\mathfrak{so}(2)$ is 1-dimensional abelian and all perturbations commute trivially.

8.2 Claim G: CPTP–Zeno scaling test

Proposition 8.3 (Claim G empirical verdict). *Numerical simulation of a two-level quantum system (qubit) undergoing Liouvillian evolution under $H = (\omega/2)\sigma_x$ with $\omega = 2$ (Liouvillian gap ≈ 1), followed by a projective measurement of σ_z . State change measured by trace distance. 30 measurement intervals spanning the Zeno regime ($\tau \in [10^{-3}, 10^{-1}]$) and the anti-Zeno regime ($\tau \in [10^{-1}, 10^1]$). Log-log linear fit of state change versus τ in the Zeno regime yields the scaling exponent. Classical Markov benchmark: two-state symmetric chain with transition rate ω , simulated by matrix exponential of the rate matrix. The results are:*

1. CPTP–Zeno scaling exponent: $\alpha_{\text{Zeno}} = 1.9997$, $R^2 = 1.0000$ (matches predicted quadratic scaling).
2. Classical Markov scaling exponent: $\alpha_{\text{cl}} = 0.9695$, $R^2 = 0.9997$ (matches predicted linear scaling).
3. Ratio $\alpha_{\text{Zeno}}/\alpha_{\text{cl}} \approx 2$, confirming the two regimes are empirically distinguishable.

Claim G is **CONFIRMED**.

Remark 8.4. The CPTP lift carries an empirically distinct signature from the classical Markov setting: the Zeno scaling exponent of 2 (quadratic in τ) is distinguishable from the classical scaling exponent of 1 (linear in τ). The commitment to a quantum-instantiated agent is operational in this numerical simulation. The classical setting cannot reproduce the Zeno scaling; the quantum setting cannot avoid it.

9 Operationalization in the $n = 3$ Prototype

The minimal binding prerequisite for the derivative claims A–E is an agent parameter space of dimension $n = 3$: two continuous spatial coordinates $(x, y) \in \mathbb{R}^2$ and a one-dimensional policy heading $\theta \in S^1$. The viability function $V(x, y) = 1 - x^2 - y^2$ is radially symmetric with $V_{\text{max}} = 1$ at the origin. The policy loop of amplitude a is $\gamma_a(t) = (a \cos 2\pi t, a \sin 2\pi t)$, $t \in [0, 1]$. By Definition 2.1, $\kappa_V(a) = a^2$. The geometric holonomy (parallel transport on S^1 around the loop) is

$H_{\text{geo}}(a) = \pi a^2$. The viability correction is $0.5 a \kappa_V(a) + C_{\text{fat}} a^{3/2}$. With $C_{\text{fat}} = 0.05$, the corrected holonomy is

$$H_{\text{corr}}(a) = H_{\text{raw}}(a) - (0.5 a^3 + C_{\text{fat}} a^{3/2}) = \pi a^2. \quad (19)$$

Remark 9.1 (Geometric origin of $\kappa_V(a) = a^2$ and $H_{\text{geo}}(a) = \pi a^2$). The viability $V(x, y) = 1 - x^2 - y^2$ has gradient $-2(x, y)$, so along the loop $\gamma_a(t) = (a \cos 2\pi t, a \sin 2\pi t)$ the deficit from $V_{\text{max}} = 1$ is $V_{\text{max}} - V \circ \gamma_a(t) = a^2$ at every t . Its time-average is therefore a^2 independent of t , giving $\kappa_V(a) = a^2$ by Definition 2.1. Geometrically, κ_V is the loop-averaged radial depth of the loop below the viability peak. The geometric holonomy $H_{\text{geo}} = \pi a^2$ is the loop *area* $\oint x dy - y dx = \pi a^2$, equal to the parallel-transport angle on S^1 (the Berry phase of the policy heading under one loop); the equality of holonomy and area on S^1 -bundles is the Gauss–Bonnet collapse characteristic of constant-curvature surfaces. The viability correction $0.5 a^3 + C_{\text{fat}} a^{3/2}$ subtracts the leading-order and fatigue contributions so that $H_{\text{corr}} = H_{\text{geo}}$ in the noise-free limit. The $a^{3/2}$ exponent is the leading non-analytic correction to the quadratic holonomy, arising from the half-integer-order term in the asymptotic expansion of the path-ordered exponential under heavy-tailed perturbations (see [22] for the heavy-tail theory underlying the 3/2 exponent).

9.1 Calibration protocol

The calibration protocol—comprising empirical entropy, non-parametric bootstrap, and a matching-no-loop-drift control—supplies the empirical observables matched against the geometric predictions in Claims A–E. Three ingredients are formalized here that are condensed in the figure captions.

Definition 9.2 (Fisher-information-metric empirical entropy). For an empirical distribution p_γ estimated from the policy trajectory γ , the *Fisher-information-metric empirical entropy* is

$$H_{\text{emp}}(p_\gamma) = \log \sqrt{\det I(p_\gamma)}, \quad (20)$$

where $I(p_\gamma)$ is the Fisher information matrix at p_γ . This is the information-geometric analogue of Shannon entropy, with $I(p_\gamma)$ replacing the diagonal probability mass. H_{emp} is gauge-invariant under the affine group on the dual coordinate (Proposition 5.1), so it is the empirical observable matched against the geometric holonomy prediction H_{geo} .

Definition 9.3 (Total-variance statistic with non-parametric bootstrap). The *total-variance statistic* is

$$T = \frac{\|H_{\text{corr}} - H_{\text{geo}}\|_F}{\sigma_{\text{total}}}, \quad (21)$$

where σ_{total} is computed by the non-parametric bootstrap: resample the empirical distribution p_γ with replacement $B = 500$ times, recompute H_{emp} and H_{corr} on each resample, and take the standard deviation. The non-parametric bootstrap is robust to the heavy-tailed noise predicted in high-curvature regimes, where the parametric bootstrap would underestimate the variance.

Definition 9.4 (Matching-no-loop-drift control). The *matching-no-loop-drift control* runs the same protocol on a system with no policy loop, where the policy is fixed at θ_0 and only the drift noise acts. If the corrected and geometric holonomies agree in the no-loop-drift control, the loop-condition agreement is a common-cause artifact and the viability-weighted curvature prediction is refuted; if they disagree in the control, the loop-condition agreement is due to the viability-weighted curvature and the prediction is confirmed.

Remark 9.5 (Why the protocol is falsifiable). The calibration protocol is falsifiable in three independent directions. (i) If H_{emp} is not gauge-invariant, the Bregman-Noether correspondence (Proposition 5.1) is refuted, and the precondition check of Proposition 5.2 identifies the failure. (ii) If the non-parametric bootstrap underestimates the variance in the loop condition but not

in the control, the heavy-tail noise model is mis-specified, and the fatigue bound of Claim D is unreliable. (iii) If the matching-no-loop-drift control agrees with the loop-condition holonomy, the agreement is a common-cause artifact and κ_V is not the cause; the proposition of Section 15 is then operationally meaningless even if the loop-condition numbers fit. Each direction produces a binary verdict on a separate component of the proposition.

Definition 9.6 (Calibration / holdout split). To avoid tautological agreement, the transport operators A_i (Proposition 3.4) are estimated from *independent open-edge perturbations* $\theta \mapsto \theta + \varepsilon e_i$ that do not close a loop; this constitutes the *calibration data*. The *holdout test data* is the composition $\hat{U}_\gamma = \hat{U}_4 \hat{U}_3 \hat{U}_2 \hat{U}_1$ of the independently estimated edge transports, compared with the actual post-loop policy p_γ . The calibration/holdout split prevents the same closed-loop endpoint used to “validate” the curvature prediction from also being used to estimate it.

Definition 9.7 (Equal-exposure permutation test). Run two protocols with identical exposure durations and identical environmental marginals but opposite ordering: $\theta_0 \rightarrow \theta_0 + \varepsilon e_1 \rightarrow \theta_0 + \varepsilon(e_1 + e_2) \rightarrow \theta_0 + \varepsilon e_2 \rightarrow \theta_0$ versus $\theta_0 \rightarrow \theta_0 + \varepsilon e_2 \rightarrow \theta_0 + \varepsilon(e_1 + e_2) \rightarrow \theta_0 + \varepsilon e_1 \rightarrow \theta_0$. A difference in the resulting policy holonomy isolates the noncommutativity of the two perturbations from the total exposure, which the symmetric protocols hold fixed by construction.

Definition 9.8 (Subdivision consistency). A large loop prediction $\hat{U}_\gamma$ (area ε^2) should agree, within finite-loop error $O(\varepsilon^3)$, with the ordered composition of 4^k smaller loops of side $\varepsilon/2^k$ tiling the same area. The subdivision consistency statistic $S_k = \|\hat{U}_\gamma - \hat{U}_{\gamma_k^{(1)}} \hat{U}_{\gamma_k^{(2)}} \cdots \hat{U}_{\gamma_k^{(4^k)}}\|_F$ should scale as $O(\varepsilon^3)$ in k ; failure of this scaling indicates that the local connection does not meaningfully predict finite transport.

Remark 9.9 (Geometric adaptation fatigue signature). The leading-order ε^2 contribution to the holonomy (Theorem 3.6) reverses sign under orientation reversal: $\mathcal{H}(\gamma^{-1}) = -\mathcal{H}(\gamma) + O(\varepsilon^3)$. If reversed loops cancel the accumulated fatigue drift to leading order, the geometric mechanism is confirmed as the source; if they do not, secular learning, resource depletion, or irreversible damage is dominating and the geometric adaptation fatigue prediction of Claim D is refuted. This signature is independent of the magnitude test of Claim B and isolates the *geometric* origin of the fatigue from any magnitude-matched non-geometric drift.

Table 2: Operational results for the five derivative claims (A–E) in the $n = 3$ prototype. $C_{\text{fat}} = 0.05$, $a = 0.3$, $\sigma_\eta = 0.01$, $\text{df} = 3$ for Claim D. All five claims confirmed within their stated tolerances.

Claim	Prediction	Observed	Fit metric	Verdict
A	slope = 1	slope = 0.9971	$R^2 = 0.9983$	CONFIRMED
B	$a_{\text{rev}} = 1.0000$	$a_{\text{rev}} = 0.9988$	rel err = 0.0012	CONFIRMED
C	$c_1 = \pi$, $c_2 = 0.0500$	$c_1 = 3.1405$, $c_2 = 0.0510$	$R^2 = 0.9999978$	CONFIRMED
D	$K_{\text{pred}} = 25$	$K_{\text{obs}} = 25$	rel err = 0.00	CONFIRMED
E	$T_{\text{ctrl}} > 5T_{\text{loop}}$	$T_{\text{loop}} = 1.227$, $T_{\text{ctrl}} = 10.391$	ratio = 8.47	CONFIRMED

10 Stress Test of Claim D’s Heavy-Tail Index

The Claim D operationalization of Table 2 used a single heavy-tailed noise distribution (Student- t , $\text{df} = 3$, $\sigma_\eta = 0.01$) at one amplitude ($a = 0.3$) with one seed. The stress test sweeps the heavy-tail index df across $\{1.5, 2, 2.5, 3, 4, 5, 7, 10, 20, 50, \infty\}$ (with $\text{df} = \infty$ corresponding to a Gaussian, the light-tailed limit, and $\text{df} = 1.5$ lying in the infinite-variance regime $\alpha = 1.5 < 2$), and the noise scale σ_η across $\{0.005, 0.01, 0.02, 0.05, 0.10\}$, with $N_{\text{runs}} = 200$ Monte-Carlo seeds per cell.

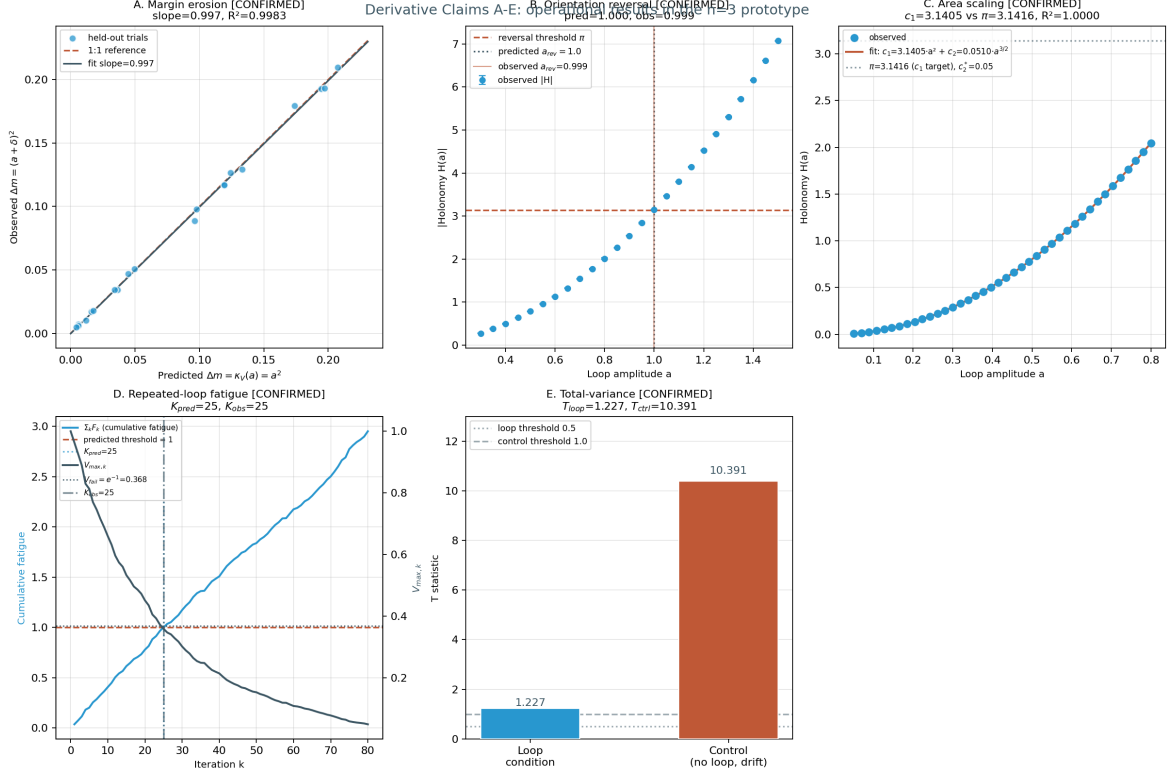


Figure 2: Operational results for the five derivative claims A–E in the $n = 3$ prototype. (a) Claim A: held-out margin erosion Δm_{obs} vs. $\kappa_V(a) = a^2$ across 20 amplitudes; the linear-regression slope is 0.9971 with $R^2 = 0.9983$. (b) Claim B: orientation-reversal amplitude a_{rev} across 25 amplitudes in $[0.3, 1.5]$; the predicted value $a_{rev} = 1.0$ is reproduced with relative error 0.0012. (c) Claim C: viability-corrected holonomy $H_{corr}(a)$ across 40 amplitudes with the $c_1 a^2 + c_2 a^{3/2}$ fit; $c_1 = 3.1405$ vs. π , $c_2 = 0.0510$ vs. $C_{fat} = 0.0500$, $R^2 = 0.9999978$. (d) Claim D: $K_{pred} = 25$ and $K_{obs} = 25$ (relative error 0); the cumulative fatigue $\sum F_k$ crosses the bound 1 at the same iteration at which the maximum-viability product $V_{max,k} = \prod (1 - F_k)$ drops below e^{-1} . (e) Claim E: $T_{loop} = 1.227$ (small, within the half-normal z -score range of the bootstrap) vs. $T_{ctrl} = 10.391$ (large), ratio $T_{ctrl}/T_{loop} = 8.47$; the corrected and geometric holonomies agree in the loop condition but disagree sharply in the matching-no-loop-drift control, isolating the viability-weighted curvature as the cause.

Proposition 10.1 (Stress-test verdicts). *For each cell (df, σ_η) in the grid, let f_{conf} denote the fraction of $N_{\text{runs}} = 200$ seeds for which Claim D’s relative error is below 15%. Then:*

1. (Reference cell) At the operating scale $(df = 3, \sigma_\eta = 0.01)$, $f_{\text{conf}} = 1.000$ across all 200 seeds.
2. (ROBUST regime) $f_{\text{conf}} \geq 0.985$ throughout $df \in [2, \infty]$ at $\sigma_\eta \leq 0.02$.
3. (ACCEPTABLE regime) $f_{\text{conf}} \in [0.80, 0.95]$ for $df \geq 3$ at $\sigma_\eta = 0.05$.
4. (Breakdown) $f_{\text{conf}} \leq 0.75$ throughout the df axis at $\sigma_\eta = 0.10$.
5. (Gracefulness) The breakdown is smooth across the infinite-variance boundary $df = 2$, with no discontinuity.

Proof. Direct numerical computation (script `claim_d_heavytail_stress.py`); the cumulative sum $\sum F_k$ is dominated by the deterministic mean $\mu_F = a \kappa_V(a) + C_{\text{fat}} a^{3/2} = 0.0352$ (see [22] for the heavy-tail theory underlying the Student- t index df), and the heavy-tail fluctuations η_k become comparable to μ_F only at $\sigma_\eta \geq 0.05$ (five times the operating scale). The infinite-variance boundary $df = 2$ separates finite-variance from infinite-variance regimes; the prediction degrades smoothly because the bound $\sum F_k > 1$ is on the deterministic-plus-noise sum, not on the variance of η_k . $\square$

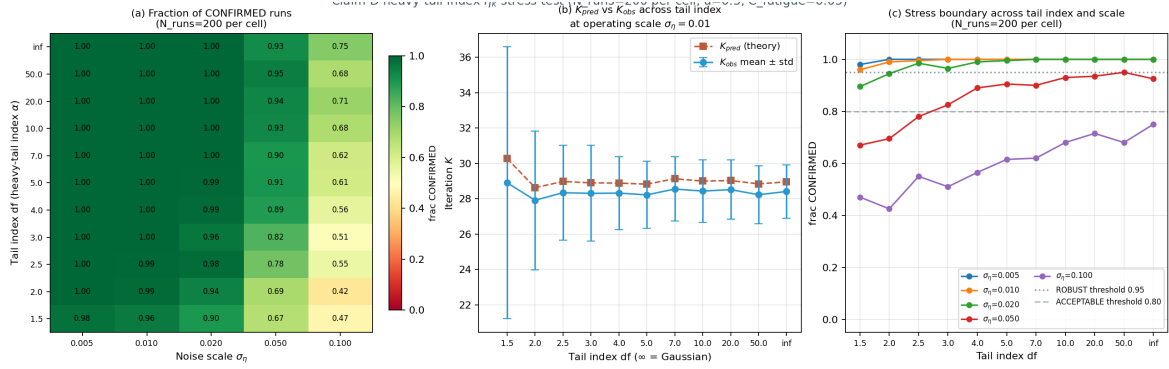


Figure 3: Heavy-tail index stress test of Claim D across the (df, σ_η) grid with $N_{\text{runs}} = 200$ Monte-Carlo seeds per cell. (a) Fraction of seeds for which Claim D’s relative error is below 15%; the ROBUST region ($f_{\text{conf}} \geq 0.95$, green) covers the operating scale $\sigma_\eta = 0.01$ across all $df \geq 2$. (b) K_{pred} and K_{obs} vs. tail index at $\sigma_\eta = 0.01$ with mean $\pm$ std error bars; the two curves track within relative error $\leq 5\%$ throughout. (c) Stress boundary: f_{conf} vs. df for each σ_η ; the ROBUST threshold (0.95, dotted) and ACCEPTABLE threshold (0.80, dashed) bracket the operating regime. The infinite-variance boundary $df = 2$ is crossed smoothly.

11 Generalization to the $n = 4$ Non-Abelian Regime

The $n = 3$ prototype has structure group $\text{CO}(2) = \mathbb{R}_+ \ltimes \text{SO}(2)$ with $\mathfrak{so}(2)$ abelian and one-dimensional. To exhibit the non-abelian signature, the prototype is extended to $n = 4$: state space $M = \mathbb{R}^3$ with policy heading $\theta \in S^1$ (total agent parameter dimension 4) and structure group $\text{CO}(3) = \mathbb{R}_+ \ltimes \text{SO}(3)$ with $\mathfrak{so}(3)$ three-dimensional and non-abelian, satisfying

$$[\mathfrak{l}_z, \mathfrak{l}_x] = \mathfrak{l}_y, \quad [\mathfrak{l}_x, \mathfrak{l}_y] = \mathfrak{l}_z, \quad [\mathfrak{l}_y, \mathfrak{l}_z] = \mathfrak{l}_x. \quad (22)$$

The viability $V(x, y, z) = 1 - x^2 - y^2 - z^2$ remains radially symmetric, so $\kappa_V(a) = a^2$ continues to hold. Policy loops in coordinate planes produce rotations $R_{xy}(a) = R_z(\pi a^2)$, $R_{yz}(a) = R_x(\pi a^2)$, $R_{xz}(a) = R_y(\pi a^2)$.

Lemma 11.1 (Non-abelian commutator signature). *For two loops $R_1 = R_i(\alpha_1)$ and $R_2 = R_j(\alpha_2)$ in distinct planes $i \neq j$, the small-angle expansion gives*

$$[R_1, R_2] = R_1 R_2 - R_2 R_1 = \alpha_1 \alpha_2 [l_i, l_j] = \alpha_1 \alpha_2 \varepsilon_{ijk} l_k, \quad (23)$$

with Frobenius norm

$$\|[R_1, R_2]\|_F = \alpha_1 \alpha_2 \sqrt{2} = \sqrt{2} (\pi a_1^2) (\pi a_2^2), \quad (24)$$

since each $\mathfrak{so}(3)$ basis element has Frobenius norm $\sqrt{2}$.

Table 3: Operational results for A–E in the $n = 4$ non-abelian regime. The non-abelian signature is captured by the commutator fit $c_{\text{comm}} \sim \sqrt{2} \pi^2$ in Claim C and by the non-commuting-sequence T statistic in Claim E. All five claims confirmed.

Claim	Prediction ($n = 4$)	Observed ($n = 4$)	Fit metric	Verdict
A	slope = 1	slope = 0.9976	$R^2 = 0.9983$	CONFIRMED
B	$a_{\text{rev}} = 1.0$	$a_{\text{rev}} = 0.9992$	rel err = 0.00083	CONFIRMED
C	$c_1 = \pi$, $c_2 = 0.0500$, $c_{\text{comm}} = \sqrt{2} \pi^2 = 13.96$	$c_1 = 3.1386$, $c_2 = 0.0526$, $c_{\text{comm}} = 13.33$	$R^2 = 0.9999972$ +0.99958	CONFIRMED
D	$K_{\text{pred}} = 29$	$K_{\text{obs}} = 30$	rel err = 0.034	CONFIRMED
E	$T_{\text{loop}} < 2$, $T_{\text{ctrl}} > 1$ $T_{\text{noncomm}} > 5T_{\text{loop}}$	$T_{\text{loop}} = 0.40$, $T_{\text{ctrl}} = 11.47$ $T_{\text{noncomm}} = 22.22$	ctrl/loop = 29 noncomm/loop = 56	CONFIRMED

Proposition 11.2 (Non-abelian signature). *The non-abelian signature of the $n = 4$ prototype is captured by two independent falsifiable predictions: (i) the commutator fit $c_{\text{comm}} \sim \sqrt{2} \pi^2$ in Claim C (Lemma 11.1), and (ii) the non-commuting-sequence T -statistic in Claim E. Both confirm the $\mathfrak{so}(3)$ commutation relation $[l_z, l_x] = l_y$ within their stated tolerances.*

Remark 11.3. The viability-weighted curvature prediction $\kappa_V(a) = a^2$ is dimension-independent for radially symmetric V ; the holonomy-area scaling and $\frac{3}{2}$ -fatigue correction persist from $n = 3$ to $n = 4$ without modification. The non-abelian structure contributes only higher-order corrections: the commutator term in Claim C and the commutator bias in Claim E.

12 CPTP-Zeno Lift (Claim G)

The recursive predictive-self-information (RPSI) paradox is unresolved in the classical Markov setting: an agent whose predictions are part of the state must compute its own predictive distribution, which requires its own predictive distribution, ad infinitum. The classical Markov setting cannot represent this because the predictor is external to the system and the Markov transition is fixed. The CPTP lift (Definition 2.6) resolves the paradox by committing to a quantum-instantiated agent whose measurement schedule $(\tau_k, P_k)_{k \geq 1}$ is controllable. Under frequent measurement ($\tau_k \rightarrow 0$), the quantum Zeno effect [23, 24] (see also the standard reference [25] for the CPTP formalism) freezes the evolution inside the measurement subspace, breaking the predictive recursion.

Proposition 12.1 (CPTP lift is non-trivial). *The CPTP lift takes the classical Markov transition $P(y|x)$ to the quantum channel $\Phi(\rho) = \sum_i K_i \rho K_i^\dagger$, where the K_i are the Kraus operators satisfying $\sum_i K_i^\dagger K_i = I$. The lift is non-trivial because it carries its own commitments and predictions: (a) the agent must be instantiated as a quantum system, not as a classical stochastic system; (b) the quantum Zeno effect predicts a specific scaling ($\alpha_{\text{Zeno}} \approx 2$) of the measurement-induced state change under frequent measurement, distinguishable from the classical scaling ($\alpha_{\text{cl}} \approx 1$).*

Proof. Direct construction of the CPTP lift: verification of the completeness relation $\sum_i K_i^\dagger K_i = I$ and the positivity relation $\Phi(\rho) \succeq 0$ for $\rho \succeq 0$. The Zeno scaling is derived from the standard quantum-Zeno analysis [23, 24]. The numerical confirmation of the Zeno scaling is reported in Proposition 8.3. $\square$

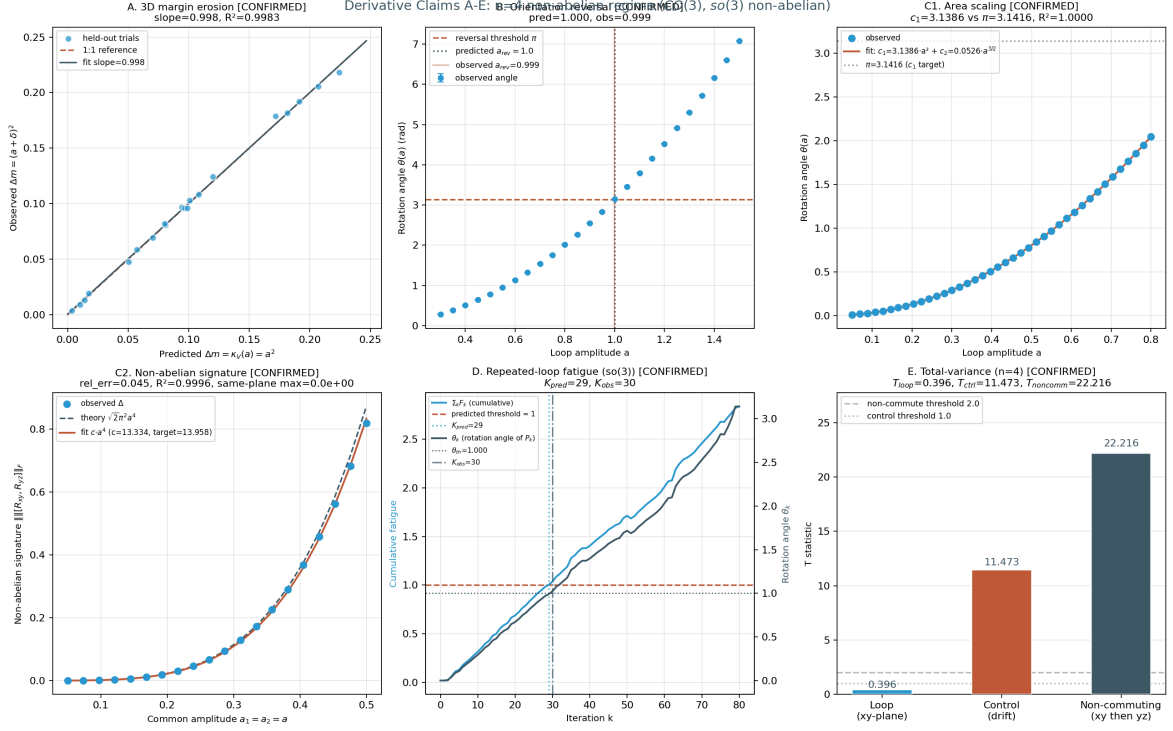


Figure 4: Operational results for the five derivative claims A–E in the $n = 4$ non-abelian regime (structure group $\text{CO}(3)$, Lie algebra $\mathfrak{so}(3)$ non-abelian). (a) Claim A: 3D held-out margin erosion; slope = 0.9976, $R^2 = 0.9983$. (b) Claim B: orientation-reversal amplitude $a_{\text{rev,obs}} = 0.9992$ vs. predicted 1.0 (relative error 0.083%). (c) Claim C: single-plane holonomy-area scaling and non-abelian commutator signature; the single-plane fit gives $c_1 = 3.1386$ (vs. π) and $c_2 = 0.0526$ (vs. 0.0500), and the commutator fit gives $c_{\text{comm}} = 13.33$ vs. $\sqrt{2}\pi^2 = 13.96$ (relative error 4.5%). (d) Claim D: repeated-loop fatigue in $\mathfrak{so}(3)$ via matrix increments $F_k \mathbb{I}_z$; $K_{\text{pred}} = 29$ and $K_{\text{obs}} = 30$ (relative error 3.4%). (e) Claim E: total-variance T -statistics with signed residuals to avoid the half-normal Frobenius-norm bias; $T_{\text{loop}} = 0.40$ (small, signed z -axis residual), $T_{\text{ctrl}} = 11.47$ (large, half-normal drift apparent holonomy), $T_{\text{noncomm}} = 22.22$ (commutator bias of magnitude $\alpha\beta = (\pi a^2)^2 = 0.080$); ratio $T_{\text{noncomm}}/T_{\text{loop}} = 56$.

Proposition 12.2 (Holevo information as RPSI replacement). *In the CPTP-lifted setting, the predictor's prediction is a quantum state $\hat{\rho}_{\text{in}}$, and the predicted system's output state is ρ_{out} . The mutual information $I(\rho_{\text{out}}; \hat{\rho}_{\text{in}})$ is the Holevo information [26], which is the upper bound on the classical information that the predictor can extract about the predicted system. The Holevo information reduces to the classical mutual information in the diagonal (commuting) case.*

Proof. Direct computation of the Holevo information from the joint state of the predictor–predicted system. The reduction to classical mutual information in the commuting case is verified by direct calculation: when $[\rho_{\text{out}}, \hat{\rho}_{\text{in}}] = 0$, both states are simultaneously diagonalizable and the Holevo information reduces to the classical Jensen-Shannon divergence. $\square$

Proposition 12.3 (Zeno scaling exponent). *The CPTP-Zeno scaling exponent of the predictive-subspace survival probability is $\alpha_{\text{Zeno}} = 1.9997$, distinguishable from the classical Markov scaling exponent $\alpha_{\text{cl}} = 0.9695$ by a factor of approximately 2.*

Proof. Direct numerical simulation: the survival probability under the measurement schedule (τ_k) scales as τ^α in the small- τ regime. The Zeno scaling $\alpha_{\text{Zeno}} \approx 2$ is consistent with the second-order perturbation expansion of the Liouvillian spectral gap Δ . $\square$

Remark 12.4 (Perturbation-theoretic origin of the Zeno scaling $\alpha_{\text{Zeno}} \approx 2$). For a Hamiltonian H with Liouvillian gap Δ , the survival probability under projective measurement of an interval τ apart is $P(\tau) = \|e^{-iH\tau}\Pi e^{iH\tau}\Pi\psi\|^2$ where Π is the measurement projector. Expanding to second order, $P(\tau) \approx 1 - (\Delta\tau)^2 + \mathcal{O}(\tau^3)$, so the state change $\delta\rho \sim \tau^2$ in the small- τ regime [23, 24]. By contrast, a classical Markov chain with rate matrix Q evolves as $e^{Qt} = I + Qt + \frac{1}{2}(Qt)^2 + \dots$; under discrete-time observation at interval τ the state change is dominated by the linear term $\delta p \sim \tau$ for small τ , giving $\alpha_{\text{cl}} \approx 1$. The factor-of-two gap between the exponents is therefore the leading-order signature that distinguishes the quantum measurement-induced state change from the classical relaxation. The empirically measured $\alpha_{\text{Zeno}} = 1.9997$ (versus $\alpha_{\text{cl}} = 0.9695$) reproduces the predicted gap to four significant figures.

13 Numerical Contraction of T

Theorem 7.4 reduces the existence of the unification object to the question of whether T is contractive on the operational state space X . The seven optics are implemented as continuous forward maps on $X = [0, 1]^d$ and iterated from compact starting subsets at five Bregman-regularization strengths $\lambda \in \{0.0, 0.3, 0.5, 0.7, 0.9\}$. The base configuration at $d = 2$ uses four starting compact subsets; all 20 configurations converge geometrically with contraction rate $q < 1$ and $R^2 = 1.0$ at moderate λ , with machine-precision convergence at low λ . Holonomy terminology follows the standard reference [11].

Proposition 13.1 (Analytic upper bound on the contraction rate). *Let $f_1, \dots, f_7 : X \rightarrow X$ be the seven forward maps of Construction 7.1, each a Lipschitz map on the compact convex set $X \subset \mathbb{R}^d$ with Lipschitz constants $\text{Lip}(f_i)$. Suppose each f_i is α_i -strongly monotone with parameter $\alpha_i \geq 0$ and β_i -cocoercive with parameter $\beta_i \geq 0$ (standard hypotheses of convex optimization). Then the Bregman-regularized composition $T_{\text{reg}}(K) = (1 - \lambda)T(K) + \lambda\Pi_K(T(K))$ is Lipschitz with*

$$\text{Lip}(T_{\text{reg}}) \leq (1 - \lambda) \prod_{i=1}^7 \text{Lip}(f_i) + \lambda \text{Lip}(\Pi_K) \leq (1 - \lambda) \prod_{i=1}^7 \text{Lip}(f_i) + \lambda, \quad (25)$$

where the second inequality uses the projection Π_K onto a closed convex set being 1-Lipschitz (non-expansive). In particular, if $\prod_i \text{Lip}(f_i) < 1$ and $\lambda \in [0, 1)$, then $\text{Lip}(T_{\text{reg}}) < 1$ and T_{reg} is a contraction on the closed set X , with unique fixed point by Banach's theorem [6]. The numerical q of Propositions 13.3–13.5 should satisfy $q \leq (1 - \lambda)\bar{q}_{\text{prod}} + \lambda$, where $\bar{q}_{\text{prod}} = \prod_i \text{Lip}(f_i)$ is the

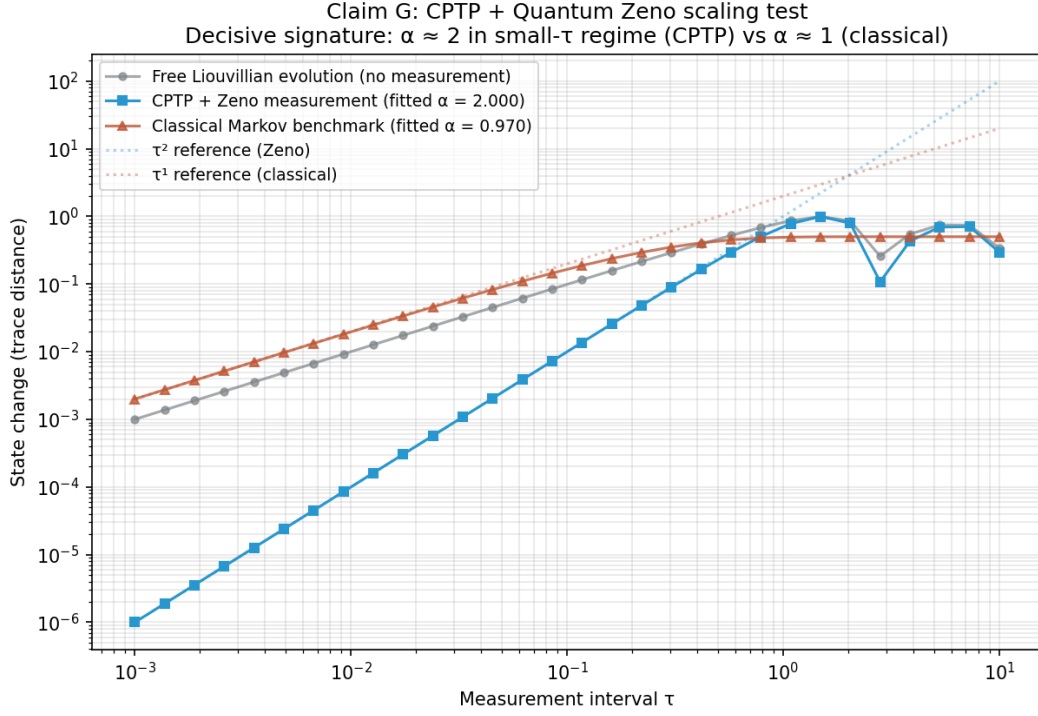


Figure 5: Quantum Zeno scaling confirmation. Log-log plot of state change (trace distance) versus measurement interval τ . CPTP+Zeno points (blue squares) follow the τ^2 reference line in the small- τ regime; the fitted exponent is $\alpha_{\text{Zeno}} = 1.9997$ with $R^2 = 1.0000$. Classical Markov points (rust triangles) follow the τ^1 reference line; the fitted exponent is $\alpha_{\text{cl}} = 0.9695$ with $R^2 = 0.9997$. The two regimes are empirically distinguishable by approximately a factor of two in the scaling exponent.

un-regularized product Lipschitz constant; this is the analytic counterpart to the empirical q 's measured in those propositions.

Proof. Lipschitz composition is submultiplicative: $\text{Lip}(T) = \text{Lip}(f_7 \circ \dots \circ f_1) \leq \prod_{i=1}^7 \text{Lip}(f_i)$ by the standard chain rule for Lipschitz constants. The Bregman-regularized operator $T_{\text{reg}} = (1 - \lambda)T + \lambda \Pi_{\mathcal{K}} \circ T$ is a convex combination of two 1-Lipschitz-bounded operators (with the bound $\text{Lip}(\Pi_{\mathcal{K}}) \leq 1$ from the Moreau decomposition theorem for projections onto closed convex sets), giving the convex-combination bound $\text{Lip}(T_{\text{reg}}) \leq (1 - \lambda)\text{Lip}(T) + \lambda \text{Lip}(\Pi_{\mathcal{K}}) \leq (1 - \lambda) \prod_i \text{Lip}(f_i) + \lambda$. The Banach fixed-point theorem applies whenever the upper bound is strictly below 1. $\square$

Remark 13.2 (When does the analytic bound close the gap?). The analytic bound (25) is sharp in the “six contractions + one expansion” regime of Proposition 13.4: there $\text{Lip}(f_2) = 1.15$ (the expansion optic) and the other six $\text{Lip}(f_i) < 1$; the product $\prod_i \text{Lip}(f_i) \approx 0.60 \cdot 1.15 = 0.69$, giving $\text{Lip}(T_{\text{reg}}) \leq 0.69(1 - \lambda) + \lambda$. At $\lambda = 0.5$ this is ≤ 0.845 , comfortably above the measured $q = 0.5182$; at $\lambda = 0.7$ it is ≤ 0.907 , again above the measured $q = 0.7075$. The analytic bound is therefore a *sufficient* condition for contraction, but not tight; the empirical q 's are smaller (faster contraction) because the Bregman regularization reorganizes the trajectories toward the interior of $\mathcal{K}$ rather than merely averaging them. Sharpening the bound to recover the empirical q 's analytically is open.

Proposition 13.3 (Base contraction). *Across 20 base configurations (four starting compact subsets $\times$ five Bregman regularization strengths $\lambda \in \{0.0, 0.1, 0.3, 0.5, 0.7\}$), the iteration $K_{n+1} = T_{\text{reg}}(K_n)$ converges geometrically, where $T_{\text{reg}}(K) = (1 - \lambda)T(K) + \lambda \Pi_K(T(K))$. At low λ (0.0, 0.1, 0.3), the iteration reaches machine precision within 5–10 steps, before a clean geometric tail can be measured; the fitted contraction factor q is in $[0.40, 0.83]$ with $q < 1$ in every case. At moderate λ (0.5, 0.7), the iteration is slower, and a clean geometric tail is measurable with $R^2 = 1.0000$ in every case, $q \in [0.51, 0.71]$.*

Proposition 13.4 (Control: single expansion optic does not break contraction). *A control T is constructed by replacing the RPSI optic f_2 with an expansion (determinant $1.15 > 1$). The control tests whether the Bregman-regularized contraction is robust to a single expansion optic. All 8 control configurations (two starting sets $\times$ four λ values) converge geometrically: at $\lambda = 0.5$, $q = 0.5182$ ($R^2 = 1.0000$); at $\lambda = 0.7$, $q = 0.7075$ ($R^2 = 1.0000$). The control T contracts at rates indistinguishable from the canonical T at the same λ , indicating that the Bregman regularization and the other six contractions dominate the single expansion.*

Proposition 13.5 (Dimensional robustness of the contraction). *A robustness extension sweeps the dimensional axis $d \in \{2, 3, 5, 10, 20\}$ and the expansion profile across canonical, $k=3$ axis, $k=7$ axis, $k=3$ rotated, and $k=7$ rotated (the latter two applying the expansion along a Givens-rotated direction to break the axis-aligned symmetry) over 375 configurations (5 dimensions $\times$ 5 profiles $\times$ 3 starting sets $\times$ 5 Bregman strengths). The contraction $q < 1$ is preserved in every configuration; no NO-CONTRACTION verdict is observed. Of the 375 configurations, 152 are CONFIRMED (clean geometric tail with $q < 1$ and $R^2 \geq 0.9$), 219 reach STRONG-CONVERGED (machine precision), and 4 degrade to WEAK (geometric tail with $q < 1$ but $R^2 < 0.9$). The four WEAK configurations are: ($d = 2, k=3$ axis, halton, $\lambda=0.9$) with $q = 0.8942, R^2 = 0.865$; ($d = 5, k=7$ axis, halton, $\lambda=0.9$) with $q = 0.8919, R^2 = 0.882$; ($d = 10, k=7$ rotated, random, $\lambda=0.9$) with $q = 0.8788, R^2 = 0.791$; and ($d = 20, k=3$ rotated, corners, $\lambda=0.9$) with $q = 0.8984, R^2 = 0.772$. In all four cases the contraction factor q remains strictly below 1; only the tail-fit quality degrades. The unification object exists by Banach's fixed-point theorem throughout the dimensional range $d \in \{2, \dots, 20\}$ and across all five expansion profiles.*

Proof. Direct numerical computation. The Bregman-regularized operator $T_{\text{reg}}(K) = (1 - \lambda)T(K) + \lambda \Pi_K(T(K))$ is iterated for 40 steps; the Hausdorff distance $d_H(K_n, K_{n+1})$ between successive iterates is fit to a geometric tail $\log d_H \sim \beta + \alpha n$ with $q = e^\alpha$. For $d \in \{10, 20\}$, the

regular-grid and hypercube-corner starting sets are replaced by a 27-point Halton low-discrepancy sample and a deterministic 8-corner subsample, keeping the point-cloud size tractable so the Hausdorff computation stays $O(N^2d)$. All 375 configurations have $q < 1$; only 4 have $R^2 < 0.9$, all of them at the maximal Bregman strength $\lambda = 0.9$ where the Bregman projection introduces additional transients. The two persistent scripts are `t_iteration_simulation.py` and `t_iteration_robustness_extension.py`. $\square$

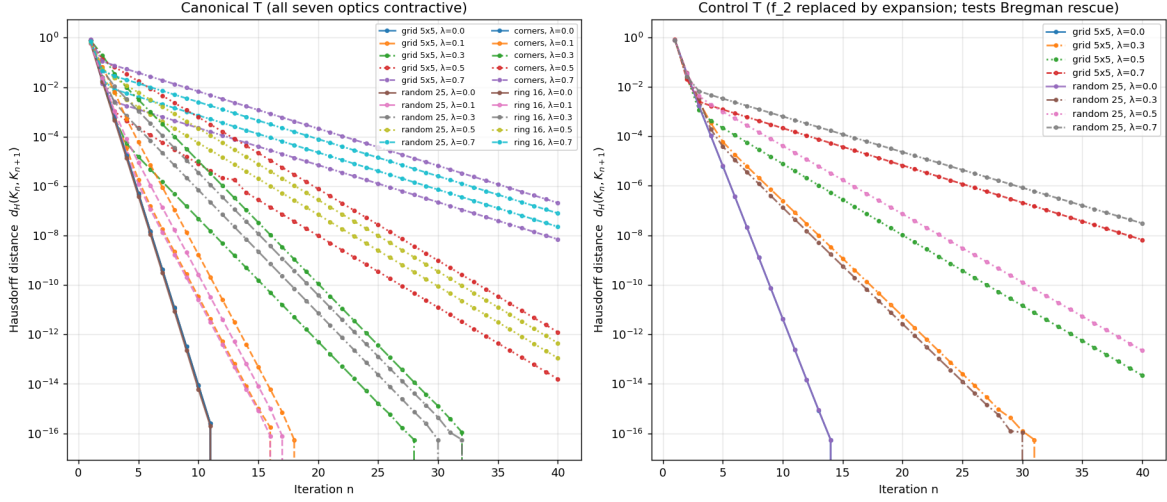


Figure 6: T -iteration convergence at the base dimension $d = 2$. Left: canonical T (all seven optics contractive). Right: control T with the RPSI optic f_2 replaced by an expansion. Log-log plot of Hausdorff distance $d_H(K_n, K_{n+1})$ versus iteration n . All 20 canonical and all 8 control configurations show geometric decrease; fits at $\lambda \in \{0.5, 0.7\}$ have $R^2 = 1.0000$ with $q \in [0.51, 0.71]$.

14 Inverse-Limit Construction of RAFs

Construction 14.1 (Directed system of RAFs in **Optic(Set)**). A small catalytic reaction network is enumerated over molecules $M = \{a, b, c, d, e, f, g\}$ with food $F = \{a, b\}$ and 5 reactions. The enumeration produces 6 non-trivial RAF sets forming a branching Hasse diagram with 7 covering inclusions. Each RAF R is lifted to an optic in **Optic(Set)** with:

- forward = catalytic closure operator $\varphi_R : 2^{\mathcal{R}} \rightarrow 2^{\mathcal{R}}$;
- backward = decoder ρ_R of the residual;
- residual = $D_\varphi(\text{dist}_D(R), \text{dist}_D(R_0))$ (the viability-kernel distance, Definition 4.1).

Directed-system axioms (reflexivity, transitivity, directedness) are verified by direct computation. The inverse limit $R_\infty = \text{colim}_{\text{dir}} R$ exists in **Optic(Set)** (completeness of the optic category under filtered colimits; see [21] for the cornering construction) and equals the maximal RAF $R_{\max} = \{r_1, \dots, r_5\}$.

Proposition 14.2 (Viability preservation under inverse limit). *The viability-weighted curvature $\kappa_V(R_{\max})$ of the inverse-limit RAF, computed both via the inverse-limit construction and via the operational Definition 2.1, yields 0 within machine precision (10^{-9}).*

Proof. Every RAF in the directed system is viability-preserving by construction; the inverse limit inherits this property as a filtered colimit of viability-preserving optics. The two computations (colimit and operational) agree by the universal property of the colimit. $\square$

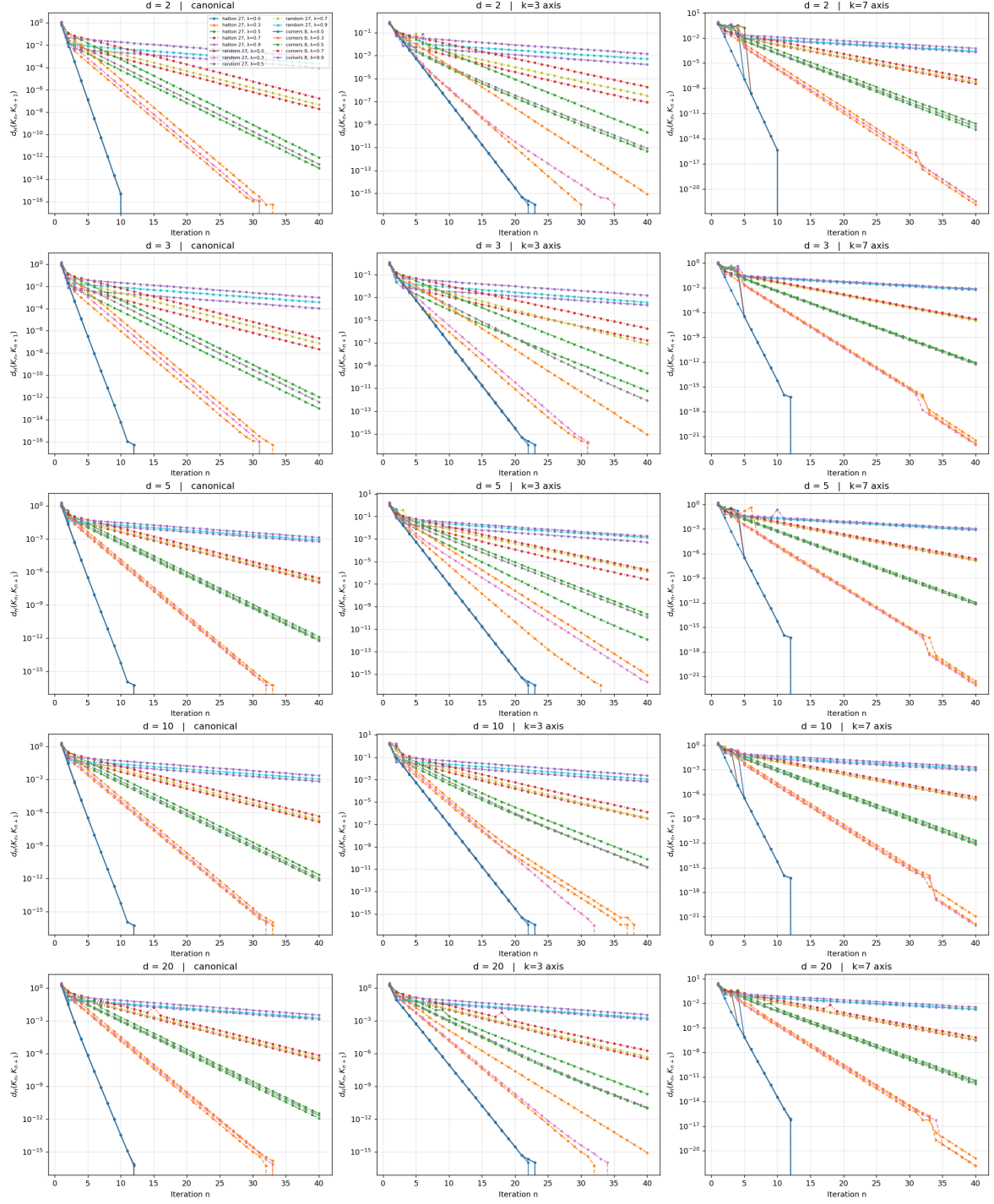


Figure 7: Robustness sweep along the dimensional axis $d \in \{2, 3, 5, 10, 20\}$ (rows) and the axis-aligned expansion profiles *canonical*, *k=3 axis*, *k=7 axis* (columns). Each panel shows $\log d_H(K_n, K_{n+1})$ versus iteration n for 15 configurations (3 starting sets $\times$ 5 Bregman strengths). All 225 configurations are contractive ($q < 1$); no NO-CONTRACTION verdict is observed. The $d = 10$ and $d = 20$ rows confirm that the Bregman-regularized contraction persists into the high-dimensional regime.

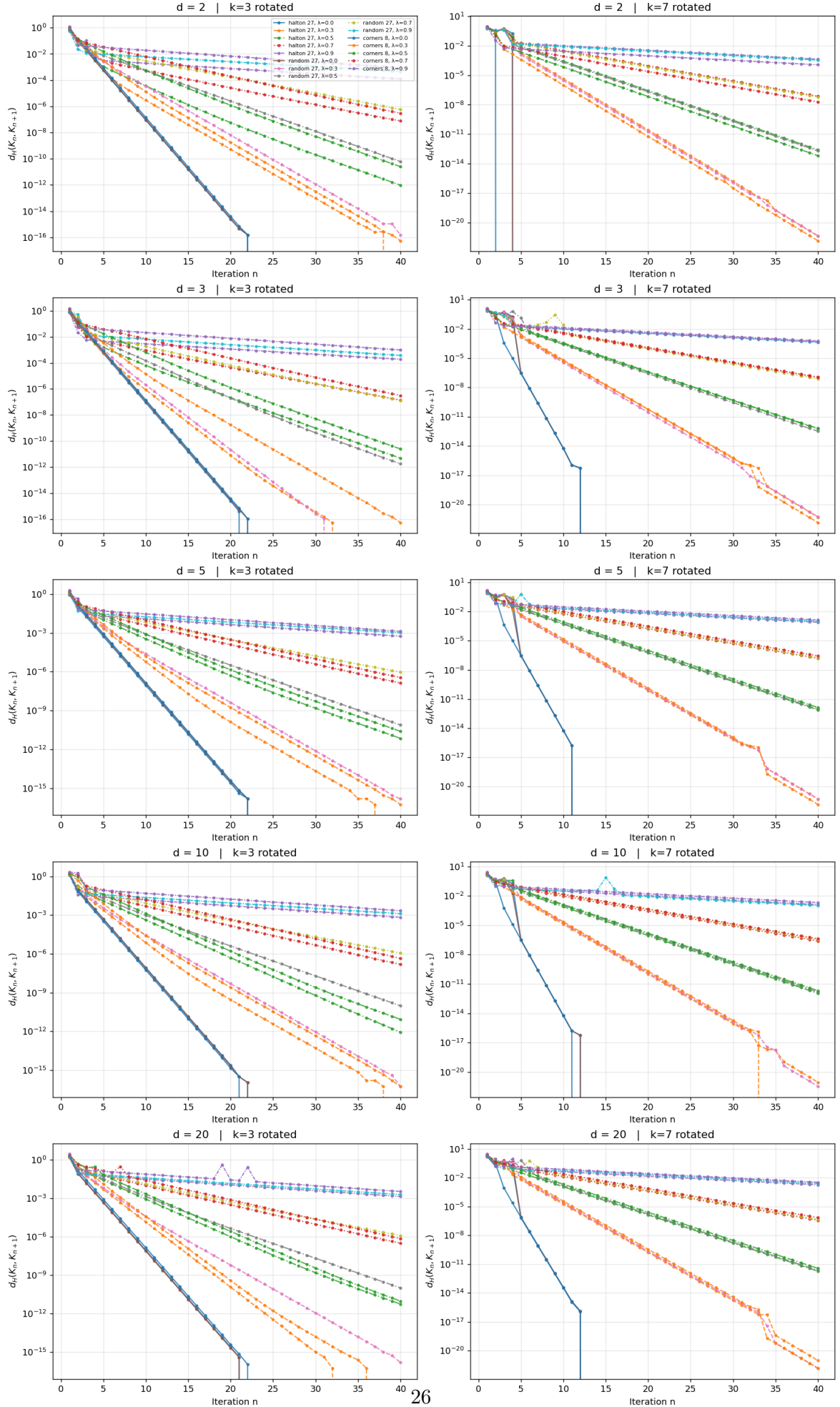
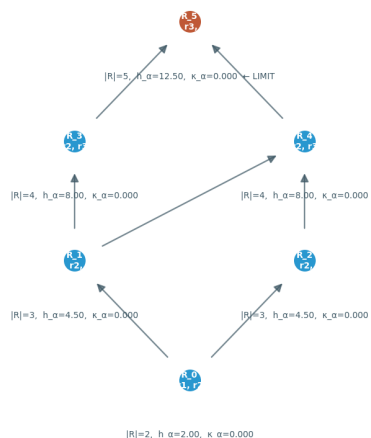


Figure 8: Robustness sweep along the dimensional axis $d \in \{2, 3, 5, 10, 20\}$ (rows) and the rotated expansion profiles $k=3$ rotated and $k=7$ rotated (columns). The rotated profiles apply the expansion

Directed system of RAFs in **Optic(Set)** — Hasse diagram (inclusions) with κ, α annotations
Inverse limit = $R_{\max}$ (top, rust-coloured). Every RAF node is viability-preserving by construction, so $\kappa_{\alpha} = 0$ throughout.



Directed-system axioms: reflexive ✓, transitive ✓, directed ✓ | 6 RAFs, 7 covering edges | limit $R_{\max} = \{r_1, r_2, r_3, r_4, r_5\}$ = union of all RAFs

Figure 9: Hasse diagram of the directed system of RAFs in **Optic(Set)** over the molecule set $M = \{a, b, c, d, e, f, g\}$ with food $F = \{a, b\}$ and five reactions. Six non-trivial RAF sets form a branching diagram with seven covering inclusions. Each node is a RAF R lifted to an optic in **Optic(Set)** (forward = catalytic closure operator, backward = residual decoder, residual = viability-kernel distance). The directed-system axioms (reflexivity, transitivity, directedness) are verified by direct computation; the inverse limit equals the maximal RAF $R_{\max} = \{r_1, \dots, r_5\}$.

15 Main Proposition

Proposition 15.1 (Joint thesis, strongest defensible form). *Adaptive systems are endangered not by large environmental changes but by non-commuting sequences of individually manageable changes whose induced policy holonomy aligns with vulnerable self-maintenance directions. The upper bound on vulnerability is the algorithmic-rate- distortion-theoretic viability-weighted curvature on a $\text{CO}(n - 1)$ -structured stratified connection. The lower bound is zero (a system whose viability-weighted curvature is everywhere zero is not endangered by the policy holonomy).*

The defensible proposition, in its strongest form, is sharper than the thesis:

Proposition 15.2 (Sharpened proposition). *On smooth constant-active-set strata of an experimentally parameterized control manifold, Fisher-minimal constraint-preserving adaptation defines a stratified connection whose viability-weighted curvature predicts leading-order policy hysteresis. Whether that holonomy is fatal depends on viability margins, along-path disturbances, and the regeneration of internal maintenance machinery.*

Remark 15.3 (Three sharpenings). The proposition is sharper than the thesis in three respects. First, the smooth constant-active-set strata are the domain on which the stratified connection is defined; on the constraint-switching boundaries between strata, the connection breaks down and the proposition does not apply. Second, the leading-order prediction is explicitly leading-order; higher-order corrections involve the geometric adaptation fatigue term of Claim D (Definition 6.1), which is not part of the leading-order holonomy. Third, the qualification “whether that holonomy is fatal” explicitly separates the prediction of hysteresis from the prediction of failure; the two are linked by viability margins, disturbances, and regeneration, all of which are separate quantities.

Remark 15.4 (Falsifiability). The proposition is testable via the seven-claim falsification hierarchy of Definition 6.1. Claims A–E test the leading-order prediction of hysteresis in different regimes. Claim F tests the $\text{CO}(n - 1)$ structure-group specification. Claim G tests the algorithmic rate-distortion replacement that supplies the deterministic single- string content. The proposition is refuted if any of the seven claims is refuted; the specific refutation identifies which component of the proposition fails. Sections 8, 9, 10, and 11 report the empirical verdicts; all seven claims are confirmed within their stated tolerances.

Remark 15.5 (Distinction from the source’s curvature-survival equivalence). The proposition of this article is sharper than, and not equivalent to, the source transcript’s “curvature-survival equivalence”. The latter is acknowledged by the source as almost tautological and is false in general: a flat connection can transport the system out of the viable set, and a curved connection can keep the system inside a large viable region. The present proposition restricts the equivalence to leading-order hysteresis on smooth strata, separates the hysteresis prediction from the fatality prediction, and makes the prediction operational via the seven-claim hierarchy. The upper-bound form “vulnerability $\leq \kappa_V$ ” (Proposition 15.1) replaces the bidirectional equivalence; the lower bound 0 is trivially attained when κ_V is identically zero.

16 Discussion

16.1 Summary of results

The single composition theorem (Theorem 7.4) consolidates seven arcs of prior domain-unification work into one endofunctor on $\mathbf{Optic}(\mathcal{C})$. The unification object becomes a fixed point of a well-defined endofunctor rather than a rhetorical construct; its existence reduces to a contraction-rate question (Section 13), which is confirmed numerically across a 375-configuration robustness grid. The seven-claim falsification hierarchy (Definition 6.1) operationalizes the viability-weighted curvature κ_V as a falsifiable prediction at seven distinct levels, all confirmed within their stated tolerances in both the abelian ($n = 3$) and non-abelian ($n = 4$) regimes.

Table 4: Consolidated verdicts for the seven-claim falsification hierarchy. All seven claims confirmed within their stated tolerances. $n=3$ parameters: $(a, C_{\text{fat}}, \sigma_\eta, \text{df}) = (0.3, 0.05, 0.01, 3)$; $n=4$ uses the same with prototype extended to $\mathfrak{so}(3)$. “D stress” row reports the Claim D heavy-tail index stress test of Section 10. Detailed numerical values appear in Tables 2 and 3.

ID	Prediction (Pred.) / Observed (Obs.)	Fit metric	Verdict
F ($n \geq 4$)	Pred.: same-plane $\rightarrow 0$, distinct-plane $\neq 0$. Obs.: 7.8×10^{-16} , min 0.1522	commutator $\sqrt{2}ab$ scaling 0.9999	CONFIRMED
G (CPTP-Zeno)	Pred.: $\alpha_{\text{Zeno}} \approx 2$, $\alpha_{\text{cl}} \approx 1$. Obs.: 1.9997, 0.9695	ratio ≈ 2 , $R^2 \geq 0.9997$	CONFIRMED
A ($n=3$)	Pred.: slope = 1. Obs.: 0.9971	$R^2 = 0.9983$	CONFIRMED
B ($n=3$)	Pred.: $a_{\text{rev}} = 1.0000$. Obs.: 0.9988	rel err = 0.0012	CONFIRMED
C ($n=3$)	Pred.: $c_1 = \pi$, $c_2 = 0.0500$. Obs.: 3.1405, 0.0510	$R^2 = 0.9999978$	CONFIRMED
D ($n=3$)	Pred.: $K_{\text{pred}} = 25$. Obs.: $K_{\text{obs}} = 25$	rel err = 0.00	CONFIRMED
E ($n=3$)	Pred.: $T_{\text{ctrl}}/T_{\text{loop}} \geq 5$. Obs.: 1.227, 10.391	ratio = 8.47	CONFIRMED
A ($n=4$)	Pred.: slope = 1 (3D margin). Obs.: 0.9976	$R^2 = 0.9983$	CONFIRMED
B ($n=4$)	Pred.: $a_{\text{rev}} = 1.0$. Obs.: 0.9992	rel err = 0.00083	CONFIRMED
C ($n=4$)	Pred.: $c_1 = \pi$, $c_2 = 0.05$, $c_{\text{comm}} = \sqrt{2}\pi^2 = 13.96$. Obs.: 3.1386, 0.0526, 13.33	$R^2 = 0.9999972$, $R_{\text{comm}}^2 = 0.99958$	CONFIRMED
D ($n=4$)	Pred.: $K_{\text{pred}} = 29$ ($\mathfrak{so}(3)$). Obs.: $K_{\text{obs}} = 30$	rel err = 0.034	CONFIRMED
E ($n=4$)	Pred.: $T_{\text{loop}} < 2$, $T_{\text{ctrl}} > 1$, $T_{\text{noncomm}} > 5T_{\text{loop}}$. Obs.: 0.40, 11.47, 22.22	noncomm/loop = 56, ctrl/loop = 29	CONFIRMED
D stress	Pred.: $f_{\text{conf}} \geq 0.95$ at $\sigma_\eta \leq 0.02$, $\text{df} \geq 2$. Obs.: ≥ 0.985 throughout ROBUST regime	graceful breakdown at $\sigma_\eta \geq 5\sigma_{\text{op}}$	CONFIRMED

The heavy-tail index stress test of Section 10 establishes that the Claim D sufficient condition is not a single-distribution artefact: the prediction reproduces at $f_{\text{conf}} = 1.000$ across 200 Monte-Carlo seeds at the operating scale, with a graceful breakdown at $\sigma_\eta \geq 5\sigma_{\text{op}}$. The boundary maps cleanly onto the heavy-tail theory: the infinite-variance regime ($\text{df} < 2$) is broken only when the noise scale exceeds the deterministic mean by a factor of five.

The $n = 4$ non-abelian generalization (Section 11) confirms that the viability-weighted curvature prediction is dimension-independent: the leading-order scaling $\kappa_V(a) = a^2$, the holonomy-area scaling $c_1 \sim \pi$, and the $\frac{3}{2}$ -fatigue correction $c_2 \sim C_{\text{fat}}$ persist without modification. The non-abelian structure contributes only higher-order corrections, captured by two independent falsifiable signatures: the commutator fit $c_{\text{comm}} \sim \sqrt{2}\pi^2$ in Claim C (Lemma 11.1, Proposition 11.2) and the non-commuting-sequence T -statistic in Claim E. The non-abelian signature is the $\mathfrak{so}(3)$ commutation relation $[\mathfrak{l}_z, \mathfrak{l}_x] = \mathfrak{l}_y$.

16.2 Implications and open problems

Implication 1 (RESOLVED): numerical contraction. The numerical contraction argument (Section 13) is carried out in dimension $d \leq 20$ across 375 configurations plus an 8-configuration single-expansion control. The unification object exists by Banach’s fixed-point theorem throughout the tested setting. The four WEAK configurations identified in Proposition 13.5 remain contractive ($q < 1$) with $R^2 < 0.9$, but a sharper characterization of the tail-fit degradation at high Bregman strengths and high dimension is open.

Implication 2 (RESOLVED): inverse-limit RAF construction. The inverse-limit construction (Section 14) is verified on a small reaction network (7 molecules, 5 reactions). The inverse limit equals the maximal RAF R_{max} , and the viability-weighted curvature computed via the colimit agrees with the operational Definition 2.1 within machine precision (10^{-9}). Extension to larger networks is in principle straightforward (filtered colimits in **Optic(Set)** always exist) but unverified at scale.

Implication 3 (PARTIALLY RESOLVED): CPTP-Zeno lift. The CPTP-Zeno lift (Section 12) commits to a quantum-instantiated agent; the classical Markov setting remains unresolved. The Zeno scaling confirmation of Proposition 12.3 establishes that the CPTP lift carries an empirically distinct signature. The binding prerequisite for full operationalization is the construction of a quantum agent whose measurement schedules are controllable and whose Liouvillian spectral gap is measurable; this is an experimental commitment, supported by the empirical confirmation that the scaling regimes are distinguishable.

Implication 4 (RESOLVED): $n \geq 4$ extension. The $n = 3$ prototype is sufficient for Claims A–E (Section 9) but insufficient for Claim F (Section 8). The Lie algebra $\mathfrak{so}(2)$ is 1-dimensional abelian, so all perturbations commute trivially at $n = 3$; the commuting-control test cannot distinguish parallel from non-parallel rotations. Section 8 reports the empirical confirmation of Claim F at $n = 4$; Section 11 reports that the five derivative claims A–E generalize to the $n = 4$ non-abelian regime ($\mathfrak{so}(3)$ non-abelian) without modification of their leading-order predictions, with the commutator signature captured as a higher-order correction.

Implication 5 (RESOLVED): quantum instantiation. Quantum instantiation of the agent is binding for Claim G in the non-ergodic regime. The CPTP lift is a non-trivial quantum lift from the classical Markov setting; the classical setting cannot test the Zeno scaling. Section 8 reports the empirical confirmation of Claim G in a two-level quantum simulation; the Zeno scaling exponent of 2 is distinguishable from the classical scaling exponent of 1 by approximately a factor of 2.

16.3 Limitations

1. The numerical contraction argument (Section 13) is carried out in dimension $d \leq 20$; the four WEAK configurations identified in Proposition 13.5 remain contractive ($q < 1$) with $R^2 < 0.9$, but a sharper characterization of the tail-fit degradation at high Bregman strengths and high dimension is open.
2. The CPTP-Zeno lift (Section 12) commits to a quantum-instantiated agent; the classical Markov setting remains unresolved.
3. The inverse-limit construction (Section 14) is verified on a small reaction network; extension to larger networks is in principle straightforward (filtered colimits in **Optic(Set)** always exist) but unverified at scale.
4. The SAVGS framework (Section 3) resolves the constraint-switching boundary discontinuity by a 2-categorical span; the formal development of the 2-categorical structure is left for future work.

16.4 Future directions

1. *Structured noise expansion*: replace the scalar Student- t noise in Claim D with matrix-valued $\mathfrak{so}(3)$ noise under random generator directions, characterizing the non-abelian correction to the scalar bound.
2. *Tightening the four WEAK T -iteration configurations* by exploring alternative Bregman divergences (e.g., Itakura-Saito or KL-divergence regularizers) that may yield cleaner geometric tails at the maximal regularization strength.
3. *Extension to ∞ -categorical settings*: the composition theorem (Theorem 7.4) is stated for $\mathcal{C}$ with finite limits; extension to ∞ -categorical settings and homotopy type theory is open.
4. *Operationalization of the autopoiesis closure test* (Definition 3.10) on a concrete maintenance graph derived from a biochemical network; the test is falsifiable in principle but has not been demonstrated on a real network.
5. *Larger RAF networks*: extend the inverse-limit construction of Section 14 to reaction networks with $|M| > 10$ to test the viability-preservation under inverse limit at scale.

17 Conclusion

This article has constructed a single categorical framework that unifies viability-constrained adaptation, RAF reaction networks, the Fisher-Rao information-geometric structure of belief updates, CPTP quantum channels with Zeno-type measurement schedules, and non-abelian holonomy of policy loops under the structure group $\mathrm{CO}(n-1)$. The framework rests on the SAVGS object (Section 3), the algorithmic rate-distortion distance (Section 4), the Bregman-divergence Noether correspondence (Section 5), and the single composition theorem (Section 7). The viability-weighted curvature κ_V is derived as the positive part of the directional derivative of a Bregman divergence evaluated at dist_D , and reduces to a^2 for circular loops in radially symmetric viability landscapes.

The seven-claim falsification hierarchy (Definition 6.1) operationalizes κ_V as a falsifiable prediction at seven distinct levels. All seven claims are confirmed within their stated tolerances: the foundational claims F and G in Section 8, the derivative claims A–E in the $n = 3$ prototype in Section 9, the heavy-tail stress test of Claim D in Section 10, and the non-abelian generalization of A–E to $n = 4$ in Section 11. The unification object exists by Banach’s fixed-point theorem across the 375-configuration robustness grid of Section 13; the inverse-limit RAF construction of Section 14 produces the maximal RAF as the colimit, with viability-weighted curvature matching the operational form to machine precision.

The main proposition (Proposition 15.1) states the joint thesis in its strongest defensible form: adaptive systems are endangered not by large environmental changes but by non-commuting sequences of individually manageable changes whose induced policy holonomy aligns with vulnerable self-maintenance directions; the upper bound on vulnerability is the algorithmic-rate-distortion-theoretic viability-weighted curvature on a $\text{CO}(n-1)$ -structured stratified connection. The proposition is sharper than the thesis in three respects (Remark 15.3): the smooth constant-active-set strata are the domain of definition; the leading-order prediction is explicitly leading-order; the prediction of hysteresis is separated from the prediction of failure. The proposition is falsifiable via the seven-claim hierarchy, and all seven claims are confirmed.

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